# AAE 50700: Principles of Dynamics
# Problem Set 6 Report

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# (6.1) Attitude Dynamics with Solar Arrays and Reaction Wheels

Consider a satellite consisting of a rigid bus $B$, two identical rigid solar arrays $A_1$ and $A_2$, and three reaction wheels, in a torque-free environment. The system center of mass $\mathcal{C}$ is fixed in the inertial frame $\mathcal{N}$. A body-fixed frame $\mathcal{B}: \{\hat{\boldsymbol{b}}_1, \hat{\boldsymbol{b}}_2, \hat{\boldsymbol{b}}_3\}$ is attached to the bus at $\mathcal{C}$. The configuration is detailed below. Fig. 1 shows a schematic illustration of the spacecraft.

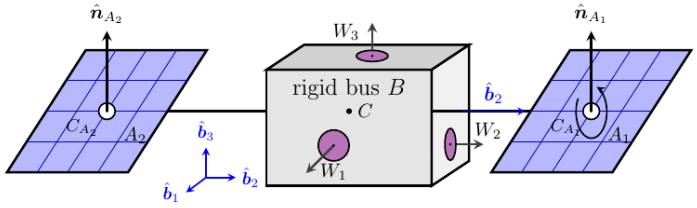



Figure 1: Schematic illustration of the spacecraft considered in PS6.

**Solar arrays:** The array centers, $C_{A_1}$ and $C_{A_2}$, are fixed relative to the bus at

$$\boldsymbol{r}_{C_{A_1}/C} = d\hat{\boldsymbol{b}}_2, \quad \boldsymbol{r}_{C_{A_2}/C} = -d\hat{\boldsymbol{b}}_2. \tag{1}$$

Each array rotates about a boom axis parallel to $\hat{\boldsymbol{b}}_2$ and passing through its own center of mass. Since the solar panels need to keep tracking the Sun, both panels have the same normal vector direction:

$$\hat{\boldsymbol{n}}_{A_1} = \hat{\boldsymbol{n}}_{A_2} = \boldsymbol{R}_2(\beta)\hat{\boldsymbol{b}}_3, \tag{2}$$

where $\boldsymbol{R}_2(\cdot)$ is a matrix representing a rotation about $\hat{\boldsymbol{b}}_2$. Assume the panel rotation angle $\beta$ is prescribed:

$$\beta(t) = \begin{cases} \beta_m \left(10\tau^3 - 15\tau^4 + 6\tau^5\right), & 0 \le t \le t_m, \\ \beta_m, & t > t_m, \end{cases} \qquad \tau := \frac{t}{t_m}. \tag{3}$$

where $t_m$, $\beta_m$ are the time and angle at the end of the maneuver, respectively, and $\tau$ is normalized time.

Denote by $\boldsymbol{I}_{A0}$ the inertia of each panel about its own center of mass when $\beta = 0$. Thus, the panel inertia with the rotation $\beta$ is:

$$\boldsymbol{I}_A(\beta) = \boldsymbol{R}_2(\beta)\boldsymbol{I}_{A0}\boldsymbol{R}_2^\top(\beta). \tag{4}$$

**Reaction wheels (RWs):** Wheel $W_i$ has a fixed spin axis $\hat{\boldsymbol{b}}_i$, axial moment of inertia $J_{w_i}$, and spin angle $\alpha_i$, for $i = 1, 2, 3$. Thus, relevant terms are defined as:

$$\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \alpha_3]^\top, \qquad \boldsymbol{\Omega} := \dot{\boldsymbol{\alpha}}, \qquad \boldsymbol{J}_w := \mathrm{diag}(J_{w_1}, J_{w_2}, J_{w_3}). \tag{5}$$

**Spacecraft bus:** Denote the bus inertia about $C$ by $\boldsymbol{I}_B$. It includes the inertia of the booms and the wheel transverse inertia components.

Express the orientation of $\mathcal{B}$ relative to $\mathcal{N}$ using (3-1-3) Euler angle sequence, defined as $\boldsymbol{\theta} = [\theta_1, \theta_2, \theta_3]^\top$. The relation between the time derivative of $\boldsymbol{\theta}$ and the angular velocity of $\mathcal{B}$ relative to $\mathcal{N}$ is given by:

$$\boldsymbol{\omega} \coloneqq {}^{\mathcal{B}}\boldsymbol{\omega}_{B/N} = \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}, \quad \boldsymbol{E}(\boldsymbol{\theta}) = \begin{bmatrix} \sin\theta_3 \sin\theta_2 & \cos\theta_3 & 0 \\ \sin\theta_2 \cos\theta_3 & -\sin\theta_3 & 0 \\ \cos\theta_2 & 0 & 1 \end{bmatrix}. \tag{6}$$

**Generalized coordinates:** Express the generalized coordinates in terms of the bus orientation and wheel spin angles:

$$\boldsymbol{q} = [\boldsymbol{\theta}^\top, \boldsymbol{\alpha}^\top]^\top. \tag{7}$$

## (a): Given

The above problem formulation.

### (a.1): Find

Discuss why it makes sense to not include $\beta(t)$ in the generalized coordinates Eq. (7).

**Analysis/Conclusions:**

> We do not include $\beta$ in the generalized coordinates because its motion is prescribed by Eq. (3). Generalized coordinates represent the degrees of freedom of the system, and since $\beta(t)$ is known for all time, it is not a free variable and not a degree of freedom. Including it would only add an equation for the drive torque required to control the panels, which is not what we want to find here. Instead, $\beta(t)$ and $\dot{\beta}(t)$ enter the Lagrangian as known functions of time through $\boldsymbol{I}_A(\beta)$ in Eq. (4) and the angular velocity of the arrays.

### (a.2): Find

Let us derive kinetic energy for each component, i.e., bus, solar array assembly, and RW assembly. Denoting them by $T_B$, $T_A$, and $T_W$, respectively, **show** that they are given by:

$$\begin{aligned} T_B &= \frac{1}{2}\boldsymbol{\omega}^\top \boldsymbol{I}_B \boldsymbol{\omega}, \quad T_A = \frac{1}{2}\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right)^\top \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right), \\ T_W &= \frac{1}{2}\left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)^\top \boldsymbol{J}_w \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) \end{aligned} \tag{8}$$

where $\boldsymbol{I}_{A/C}(\beta)$ is the combined array inertia about $C$, given by:

$$\boldsymbol{I}_{A/C}(\beta) = 2\boldsymbol{I}_A(\beta) + 2m_A d^2 \left(I_3 - \hat{\boldsymbol{b}}_2 \hat{\boldsymbol{b}}_2^\top\right) \tag{9}$$

**Analysis/Conclusions:**

It is known that the kinetic energy of a rotating body is half the moment of inertia $I$ times the angular velocity $\omega$ squared. This is generalized to three dimensions by using the moment of inertia tensor $\boldsymbol{I}$ and taking the inner product of the angular velocity vector $\boldsymbol{\omega}$:

$$T = \frac{1}{2} I \omega^2, \quad T = \frac{1}{2} \boldsymbol{\omega}^\top \boldsymbol{I} \boldsymbol{\omega} \tag{10}$$

For the equation of the kinetic energy of the bus $T_B$, we can substitute in the inertia tensor of the bus $\boldsymbol{I}_B$ and the angular velocity of the bus $\boldsymbol{\omega}$ (which happens to use the same symbol in this case). Since $\boldsymbol{\theta}$ measures the orientation of the $\mathcal{B}$-frame with respect to the $\mathcal{N}$-frame, the inertial angular velocity of the bus follows directly from Eq. (6), $\boldsymbol{\omega} = \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}$, and depends only on the generalized coordinates $\boldsymbol{\theta}$ and their rates $\dot{\boldsymbol{\theta}}$. Thus, the kinetic energy of the bus is:

$$T_B = \frac{1}{2} \dot{\boldsymbol{\theta}}^\top \boldsymbol{E}^\top(\boldsymbol{\theta}) \boldsymbol{I}_B \boldsymbol{E}(\boldsymbol{\theta}) \dot{\boldsymbol{\theta}} = \frac{1}{2} \boldsymbol{\omega}^\top \boldsymbol{I}_B \boldsymbol{\omega} \tag{11}$$

The kinetic energy of the reaction wheels $T_W$ follows a similar structure to the bus kinetic energy, but with an additional degree of rotation with respect to the $\mathcal{N}$-frame, modeled by the angles of rotation $\boldsymbol{\alpha}$, where each angle is the rotation of one of the three wheels relative to the bus. Since each wheel's spin is aligned with one of the $\mathcal{B}$-frame axes, its spin rate $\dot{\alpha}_i = \Omega_i$ simply adds to the bus angular velocity component $\omega_i$ along that same axis. So we can represent the angular rates of the reaction wheels as:

$$\boldsymbol{\omega} + \dot{\boldsymbol{\alpha}} = \boldsymbol{\omega} + \boldsymbol{\Omega} \tag{12}$$

Only the wheels' axial inertias $\boldsymbol{J}_w$ are needed here, as their transverse inertia components are already included in $\boldsymbol{I}_B$. Thus, we can substitute in $\boldsymbol{J}_w$ for the inertia tensor, and the kinetic energy of the reaction wheels is:

$$T_W = \frac{1}{2} (\boldsymbol{\omega} + \boldsymbol{\Omega})^\top \boldsymbol{J}_w (\boldsymbol{\omega} + \boldsymbol{\Omega}) \tag{13}$$

The arrays' kinetic energy follows a similar structure for the angular rate component, with some additional steps for the inertia tensor.

First, we know the solar arrays are, like the reaction wheels, mounted to the bus with their rotational axes aligned with a single axis of the body frame. The orientation of the solar panels is given in Eq. (2) as the vector normal to the surface of the two panels. Specifically, both panels are always oriented in the same direction and rotate together by angle $\beta$ about $\hat{\boldsymbol{b}}_2$. An angular rate vector always points in the direction of its axis of rotation, with magnitude equal to the rate of change of the rotation angle, so we can simply represent the angular rate of the solar panels relative to the $\mathcal{B}$-frame as:

$$\dot{\beta}\hat{\boldsymbol{b}}_2 \tag{14}$$

Then the structure follows the same idea as the reaction wheels, where the angular rate of the arrays aligns with the $\hat{\boldsymbol{b}}_2$ axis of the bus and adds directly to the bus angular velocity, giving the angular rate of the solar panels relative to the $\mathcal{N}$-frame as:

$$\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2 \tag{15}$$

Then, the inertia tensor of each array is given by $\boldsymbol{I}_A$. However, there are two arrays who are offset from the center of mass. This means we must derive the offset inertia tensor of the arrays by the parallel axis theorem, then add the two derivations together to get the inertia tensor of the combined two-array system $\boldsymbol{I}_{A/C}$.

Recall the parallel axis theorem says we can find inertia tensor displaced from a center of mass $\boldsymbol{I}_P$ by using the following expression where $\boldsymbol{d}$ is a displacement vector, and $\boldsymbol{I}_c$ is the inertia tensor of the object relative to its own center of mass:

$$\boldsymbol{I}_P = \boldsymbol{I}_c + m(\|\boldsymbol{d}\|^2 I_3 - \boldsymbol{d}\boldsymbol{d}^\top) \tag{16}$$

From the problem definition we know the displacement of the two panels are given in Eq (1) as $\boldsymbol{r}_{C_{A_1}/C}$ and $\boldsymbol{r}_{C_{A_2}/C}$. Using $\boldsymbol{I}_A$ as the $\boldsymbol{I}_c$ value, and $m_A$ to represent the mass of an array, we find the offset tensors to be:

$$\begin{aligned}
\boldsymbol{I}_{A1/C} &= \boldsymbol{I}_A + m_A\left(\|\boldsymbol{r}_{C_{A_1}/C}\|^2 I_3 - \boldsymbol{r}_{C_{A_1}/C}\boldsymbol{r}_{C_{A_1}/C}^\top\right) \\
&= \boldsymbol{I}_A + m_A\left(\|d\hat{\boldsymbol{b}}_2\|^2 I_3 - (d\hat{\boldsymbol{b}}_2)(d\hat{\boldsymbol{b}}_2)^\top\right) \\
&= \boldsymbol{I}_A + m_A\left(d^2 I_3 - d^2\hat{\boldsymbol{b}}_2\hat{\boldsymbol{b}}_2^\top\right) \\
&= \boldsymbol{I}_A + m_A d^2\left(I_3 - \hat{\boldsymbol{b}}_2\hat{\boldsymbol{b}}_2^\top\right)
\end{aligned} \tag{17}$$

And the negative sign cancels with $\boldsymbol{r}_{C_{A_2}/C}$:

$$\begin{aligned}
\boldsymbol{I}_{A2/C} &= \boldsymbol{I}_A + m_A\left(\|\boldsymbol{r}_{C_{A_2}/C}\|^2 I_3 - \boldsymbol{r}_{C_{A_2}/C}\boldsymbol{r}_{C_{A_2}/C}^\top\right) \\
&= \boldsymbol{I}_A + m_A\left(\|-d\hat{\boldsymbol{b}}_2\|^2 I_3 - (-d\hat{\boldsymbol{b}}_2)(-d\hat{\boldsymbol{b}}_2)^\top\right) \\
&= \boldsymbol{I}_A + m_A\left(d^2 I_3 - d^2\hat{\boldsymbol{b}}_2\hat{\boldsymbol{b}}_2^\top\right) \\
&= \boldsymbol{I}_A + m_A d^2\left(I_3 - \hat{\boldsymbol{b}}_2\hat{\boldsymbol{b}}_2^\top\right)
\end{aligned} \tag{18}$$

So, if we add together the two tensors to get $\boldsymbol{I}_{A/C}$ we get:

$$\begin{aligned}\boldsymbol{I}_{A/C} &= \boldsymbol{I}_{A1/C} + \boldsymbol{I}_{A2/C} \\ &= \left(\boldsymbol{I}_A + m_A d^2 \left(I_3 - \hat{\boldsymbol{b}}_2 \hat{\boldsymbol{b}}_2^\top\right)\right) + \left(\boldsymbol{I}_A + m_A d^2 \left(I_3 - \hat{\boldsymbol{b}}_2 \hat{\boldsymbol{b}}_2^\top\right)\right) \\ &= 2\boldsymbol{I}_A + 2m_A d^2 \left(I_3 - \hat{\boldsymbol{b}}_2 \hat{\boldsymbol{b}}_2^\top\right)\end{aligned} \tag{19}$$

Thus, with the angular rate found in Eq. (15) and the combined inertia tensor found in Eq. (19), we define the kinetic energy $T_A$ as:

$$\begin{aligned}T_A &= \frac{1}{2}\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right)^\top \boldsymbol{I}_{A/C}\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) \\ &= \frac{1}{2}\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right)^\top \left(2\boldsymbol{I}_A + 2m_A d^2 \left(I_3 - \hat{\boldsymbol{b}}_2 \hat{\boldsymbol{b}}_2^\top\right)\right)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right)\end{aligned} \tag{20}$$

**(a.3): Find**

Noting that the total kinetic energy is $T = T_B + T_A + T_W$, derive the Lagrangian $L$ for this system and express it in a form that explicitly depends on $(\boldsymbol{q}, \dot{\boldsymbol{q}})$ by eliminating $\boldsymbol{\omega}$, $\boldsymbol{\Omega}$ from $L$.

**Analysis/Conclusions:**

In this model, there is no potential energy $V$ present, so the Lagrangian $L$ is equivalent to the total kinetic energy $T$. So, we can begin to write the Lagrangian as:

$$\begin{aligned}L &= T_B + T_A + T_W \\ &= \frac{1}{2}\boldsymbol{\omega}^\top \boldsymbol{I}_B \boldsymbol{\omega} + \frac{1}{2}\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right)^\top \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \frac{1}{2}\left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)^\top \boldsymbol{J}_w \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) \\ &= \frac{1}{2}\left(\boldsymbol{\omega}^\top \boldsymbol{I}_B \boldsymbol{\omega} + \left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right)^\top \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)^\top \boldsymbol{J}_w \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)\right)\end{aligned} \tag{21}$$

Then to put it in the form that depends on $(\boldsymbol{q}, \dot{\boldsymbol{q}})$, the problem definition explicitly defines $\boldsymbol{\Omega}$ as equivalent to $\dot{\boldsymbol{\alpha}}$ and also relates $\boldsymbol{\omega}$ to $\boldsymbol{\theta}$ by:

$$\boldsymbol{\omega} = \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}, \quad \text{Where:} \quad \boldsymbol{E}(\boldsymbol{\theta}) = \begin{bmatrix} \sin\theta_3 \sin\theta_2 & \cos\theta_3 & 0 \\ \sin\theta_2 \cos\theta_3 & -\sin\theta_3 & 0 \\ \cos\theta_2 & 0 & 1 \end{bmatrix} \tag{22}$$

So these substitute into the Lagrangian as:

$$\begin{aligned} L = \frac{1}{2}\Bigg( & \Big(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}\Big)^{\top} \boldsymbol{I}_B \Big(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}\Big) \\ & + \Big(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\beta}\hat{\boldsymbol{b}}_2\Big)^{\top} \boldsymbol{I}_{A/C}(\beta) \Big(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\beta}\hat{\boldsymbol{b}}_2\Big) \\ & + \Big(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}}\Big)^{\top} \boldsymbol{J}_w \Big(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}}\Big) \Bigg) \end{aligned} \tag{23}$$

$$\text{Where:} \quad \boldsymbol{E}(\boldsymbol{\theta}) = \begin{bmatrix} \sin\theta_3 \sin\theta_2 & \cos\theta_3 & 0 \\ \sin\theta_2 \cos\theta_3 & -\sin\theta_3 & 0 \\ \cos\theta_2 & 0 & 1 \end{bmatrix}$$

Which gives the Lagrangian in a form that explicitly depends on $(\boldsymbol{q}, \dot{\boldsymbol{q}})$.

**(a.4): Find**

Let $u_i$ be the motor torque applied to $W_i$, and $\boldsymbol{u} = [u_1, u_2, u_3]^{\top}$. Derive the EOM using Lagrange's equations, and show the result in a vector-matrix form: $\boldsymbol{M}(\boldsymbol{q}, t)\ddot{\boldsymbol{q}} = f(\boldsymbol{q}, \dot{\boldsymbol{q}}, t, \boldsymbol{u})$.

**Analysis/Conclusions:**

Before we can derive the Lagrange equations, we must first determine the generalized non-conservative forces, which in this scenario come only from the motor torques $\boldsymbol{u}$. We relate $\boldsymbol{u}$ to $\boldsymbol{q}$ through virtual work. If we consider virtual work from the perspective of a torque times an angle, we know the torques themselves act directly on the angles $\boldsymbol{\alpha}$ since the torques are acting directly on the reaction wheels. This means that:

$$\begin{aligned} \delta W &= \boldsymbol{u}^{\top}\delta\boldsymbol{\alpha} \\ &= \Big(u_1\hat{\boldsymbol{b}}_1 + u_2\hat{\boldsymbol{b}}_2 + u_3\hat{\boldsymbol{b}}_3\Big)^{\top} \Big(\delta\alpha_1\hat{\boldsymbol{b}}_1 + \delta\alpha_2\hat{\boldsymbol{b}}_2 + \delta\alpha_3\hat{\boldsymbol{b}}_3\Big) \\ &= u_1\delta\alpha_1 + u_2\delta\alpha_2 + u_3\delta\alpha_3 \\ &= Q_{\alpha_1}\delta\alpha_1 + Q_{\alpha_2}\delta\alpha_2 + Q_{\alpha_3}\delta\alpha_3 = \boldsymbol{Q}_{\alpha}^{\top}\delta\boldsymbol{\alpha} \end{aligned} \tag{24}$$

$$\text{So: } Q_{\alpha_1} = u_1, \quad Q_{\alpha_2} = u_2, \quad Q_{\alpha_3} = u_3$$

Where $Q_{\alpha_i}$ is the generalized non-conservative force associated with each of the three $\alpha$ angles. This relates each torque directly to its own wheel angle and leaves no remaining non-conservative forces, so the non-conservative forces related to $\theta$ angles are all equal to zero.

$$Q_{\theta_1} = 0, \quad Q_{\theta_2} = 0, \quad Q_{\theta_3} = 0 \tag{25}$$

Then, we define the following quantities to make the EoM derivation via Lagrange's equation with the Lagrangian in Eq. (23) simpler.

First we define total spacecraft angular momentum about the CoM $\mathcal{C}$ as:

$$\boldsymbol{H}_S = \boldsymbol{I}_B\boldsymbol{\omega} + \boldsymbol{I}_{A/C}\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{J}_w\left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) \tag{26}$$

Then we define instantaneous total spacecraft inertia about $\mathcal{C}$ as:

$$\boldsymbol{I}_S = \boldsymbol{I}_B + \boldsymbol{I}_{A/C} + \boldsymbol{J}_w \tag{27}$$

So, we then apply Lagrange's equation to each coordinate $q_i$ in $\boldsymbol{q}$, which we group into the $\boldsymbol{\theta}$ and $\boldsymbol{\alpha}$ blocks:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = Q_i, \qquad i = 1, \ldots, 6 \tag{28}$$

Below we solve the partials individually for easier reading. Because of the usage of $\boldsymbol{\theta}$ and $\boldsymbol{\alpha}$ as vectors in the Lagrangian, the most convenient form ended up being taking the derivatives with respect to $\boldsymbol{\theta}$ and $\boldsymbol{\alpha}$ rather than component-wise:

$$\begin{aligned}\frac{\partial L}{\partial \boldsymbol{\theta}} = \frac{1}{2}\Bigg(&\frac{\partial}{\partial \boldsymbol{\theta}}\left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}\right)^\top \boldsymbol{I}_B\left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}\right)\\ &+ \frac{\partial}{\partial \boldsymbol{\theta}}\left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\beta}\hat{\boldsymbol{b}}_2\right)^\top \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\beta}\hat{\boldsymbol{b}}_2\right)\\ &+ \frac{\partial}{\partial \boldsymbol{\theta}}\left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}}\right)^\top \boldsymbol{J}_w\left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}}\right)\Bigg)\end{aligned} \tag{29}$$

Then work out the derivative of each term:

$$\begin{aligned}&= \frac{\partial}{\partial \boldsymbol{\theta}}\left(\left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}\right)^\top \boldsymbol{I}_B\left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}\right)\right)\\ &= \left(\frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}}\right)^\top \boldsymbol{I}_B\left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}\right) + \left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}\right)^\top \boldsymbol{I}_B\left(\frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}}\right)\\ &= 2\left(\frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}}\right)^\top \boldsymbol{I}_B\left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}\right)\\ &= 2\left(\frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}}\right)^\top \boldsymbol{I}_B\boldsymbol{\omega}\end{aligned} \tag{30}$$

For the array term:

$$
\begin{aligned}
&= \frac{\partial}{\partial \boldsymbol{\theta}} \left( \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\beta}\hat{\boldsymbol{b}}_2 \right)^\top \boldsymbol{I}_{A/C}(\beta) \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\beta}\hat{\boldsymbol{b}}_2 \right) \right) \\
&= \left( \frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}} \right)^\top \boldsymbol{I}_{A/C}(\beta) \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\beta}\hat{\boldsymbol{b}}_2 \right) + \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\beta}\hat{\boldsymbol{b}}_2 \right)^\top \boldsymbol{I}_{A/C}(\beta) \left( \frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}} \right) \\
&= 2 \left( \frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}} \right)^\top \boldsymbol{I}_{A/C}(\beta) \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\beta}\hat{\boldsymbol{b}}_2 \right) \\
&= 2 \left( \frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}} \right)^\top \boldsymbol{I}_{A/C}(\beta) \left( \boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2 \right)
\end{aligned}
\tag{31}
$$

For the wheel term:

$$
\begin{aligned}
&= \frac{\partial}{\partial \boldsymbol{\theta}} \left( \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}} \right)^\top \boldsymbol{J}_w \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}} \right) \right) \\
&= \left( \frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}} \right)^\top \boldsymbol{J}_w \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}} \right) + \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}} \right)^\top \boldsymbol{J}_w \left( \frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}} \right) \\
&= 2 \left( \frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}} \right)^\top \boldsymbol{J}_w \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}} \right) \\
&= 2 \left( \frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}} \right)^\top \boldsymbol{J}_w \left( \boldsymbol{\omega} + \boldsymbol{\Omega} \right)
\end{aligned}
\tag{32}
$$

Note in the above equations, we substitute $\boldsymbol{\omega}$ for $\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}$ and $\boldsymbol{\Omega}$ for $\dot{\boldsymbol{\alpha}}$ since they have been determined to be equivalent by the problem setup. So, if we substitute these components into Eq. (29):

$$
\begin{aligned}
\frac{\partial L}{\partial \boldsymbol{\theta}} &= \frac{1}{2} \Bigg( 2 \left( \frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}} \right)^\top \boldsymbol{I}_B \boldsymbol{\omega} + 2 \left( \frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}} \right)^\top \boldsymbol{I}_{A/C}(\beta) \left( \boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2 \right) \\
&\qquad + 2 \left( \frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}} \right)^\top \boldsymbol{J}_w \left( \boldsymbol{\omega} + \boldsymbol{\Omega} \right) \Bigg) \\
&= \left( \frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}} \right)^\top \left( \boldsymbol{I}_B \boldsymbol{\omega} + \boldsymbol{I}_{A/C}(\beta) \left( \boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2 \right) + \boldsymbol{J}_w \left( \boldsymbol{\omega} + \boldsymbol{\Omega} \right) \right) \\
&= \left( \frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}} \right)^\top \boldsymbol{H}_S
\end{aligned}
\tag{33}
$$

Which makes use of the earlier defined $\boldsymbol{H}_S$ term.

Then, we find the partial derivative $\frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}$. At this point, it must be done component-wise of $\boldsymbol{\theta}$.

For $\theta_1$:

$$\begin{aligned}\frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \theta_1} &= \frac{\partial}{\partial \theta_1}\begin{bmatrix}\sin\theta_3\sin\theta_2 & \cos\theta_3 & 0\\ \sin\theta_2\cos\theta_3 & -\sin\theta_3 & 0\\ \cos\theta_2 & 0 & 1\end{bmatrix}\\ &= \begin{bmatrix}\frac{\partial}{\partial\theta_1}(\sin\theta_3\sin\theta_2) & \frac{\partial}{\partial\theta_1}(\cos\theta_3) & \frac{\partial}{\partial\theta_1}(0)\\ \frac{\partial}{\partial\theta_1}(\sin\theta_2\cos\theta_3) & \frac{\partial}{\partial\theta_1}(-\sin\theta_3) & \frac{\partial}{\partial\theta_1}(0)\\ \frac{\partial}{\partial\theta_1}(\cos\theta_2) & \frac{\partial}{\partial\theta_1}(0) & \frac{\partial}{\partial\theta_1}(1)\end{bmatrix}\\ &= \begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}\end{aligned} \tag{34}$$

For $\theta_2$:

$$\begin{aligned}\frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \theta_2} &= \frac{\partial}{\partial \theta_2}\begin{bmatrix}\sin\theta_3\sin\theta_2 & \cos\theta_3 & 0\\ \sin\theta_2\cos\theta_3 & -\sin\theta_3 & 0\\ \cos\theta_2 & 0 & 1\end{bmatrix}\\ &= \begin{bmatrix}\frac{\partial}{\partial\theta_2}(\sin\theta_3\sin\theta_2) & \frac{\partial}{\partial\theta_2}(\cos\theta_3) & \frac{\partial}{\partial\theta_2}(0)\\ \frac{\partial}{\partial\theta_2}(\sin\theta_2\cos\theta_3) & \frac{\partial}{\partial\theta_2}(-\sin\theta_3) & \frac{\partial}{\partial\theta_2}(0)\\ \frac{\partial}{\partial\theta_2}(\cos\theta_2) & \frac{\partial}{\partial\theta_2}(0) & \frac{\partial}{\partial\theta_2}(1)\end{bmatrix}\\ &= \begin{bmatrix}\sin\theta_3\cos\theta_2 & 0 & 0\\ \cos\theta_2\cos\theta_3 & 0 & 0\\ -\sin\theta_2 & 0 & 0\end{bmatrix}\end{aligned} \tag{35}$$

For $\theta_3$:

$$\begin{aligned}\frac{\partial \boldsymbol{E}(\boldsymbol{\theta})}{\partial \theta_3} &= \frac{\partial}{\partial \theta_3}\begin{bmatrix}\sin\theta_3\sin\theta_2 & \cos\theta_3 & 0\\ \sin\theta_2\cos\theta_3 & -\sin\theta_3 & 0\\ \cos\theta_2 & 0 & 1\end{bmatrix}\\ &= \begin{bmatrix}\frac{\partial}{\partial\theta_3}(\sin\theta_3\sin\theta_2) & \frac{\partial}{\partial\theta_3}(\cos\theta_3) & \frac{\partial}{\partial\theta_3}(0)\\ \frac{\partial}{\partial\theta_3}(\sin\theta_2\cos\theta_3) & \frac{\partial}{\partial\theta_3}(-\sin\theta_3) & \frac{\partial}{\partial\theta_3}(0)\\ \frac{\partial}{\partial\theta_3}(\cos\theta_2) & \frac{\partial}{\partial\theta_3}(0) & \frac{\partial}{\partial\theta_3}(1)\end{bmatrix}\\ &= \begin{bmatrix}\cos\theta_3\sin\theta_2 & -\sin\theta_3 & 0\\ -\sin\theta_2\sin\theta_3 & -\cos\theta_3 & 0\\ 0 & 0 & 0\end{bmatrix}\end{aligned} \tag{36}$$

Then, we find the $\frac{\partial L}{\partial \dot{\boldsymbol{\theta}}}$ evaluation:

$$\begin{aligned}\frac{\partial L}{\partial \dot{\boldsymbol{\theta}}} = \frac{1}{2}\Bigg(&\frac{\partial}{\partial \dot{\boldsymbol{\theta}}}\left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}\right)^{\top}\boldsymbol{I}_B\left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}\right)\\ &+\frac{\partial}{\partial \dot{\boldsymbol{\theta}}}\left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}+\dot{\beta}\hat{\boldsymbol{b}}_2\right)^{\top}\boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}+\dot{\beta}\hat{\boldsymbol{b}}_2\right)\\ &+\frac{\partial}{\partial \dot{\boldsymbol{\theta}}}\left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}+\dot{\boldsymbol{\alpha}}\right)^{\top}\boldsymbol{J}_w\left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}+\dot{\boldsymbol{\alpha}}\right)\Bigg)\end{aligned} \tag{37}$$

Then work out the derivative of each term. This will set up a substitution similar to what was done for $\frac{\partial L}{\partial \boldsymbol{\theta}}$.

For the bus term:

$$\begin{aligned}
&= \frac{\partial}{\partial \dot{\boldsymbol{\theta}}} \left( \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} \right)^\top \boldsymbol{I}_B \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} \right) \right) \\
&= \left( \boldsymbol{E}(\boldsymbol{\theta}) \frac{\partial \dot{\boldsymbol{\theta}}}{\partial \dot{\boldsymbol{\theta}}} \right)^\top \boldsymbol{I}_B \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} \right) + \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} \right)^\top \boldsymbol{I}_B \left( \boldsymbol{E}(\boldsymbol{\theta}) \frac{\partial \dot{\boldsymbol{\theta}}}{\partial \dot{\boldsymbol{\theta}}} \right) \\
&= 2 \left( \boldsymbol{E}(\boldsymbol{\theta}) \frac{\partial \dot{\boldsymbol{\theta}}}{\partial \dot{\boldsymbol{\theta}}} \right)^\top \boldsymbol{I}_B \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} \right) \\
&= 2\boldsymbol{E}(\boldsymbol{\theta})^\top \boldsymbol{I}_B \boldsymbol{\omega}
\end{aligned} \tag{38}$$

For the array term:

$$\begin{aligned}
&= \frac{\partial}{\partial \dot{\boldsymbol{\theta}}} \left( \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\beta}\hat{\boldsymbol{b}}_2 \right)^\top \boldsymbol{I}_{A/C}(\beta) \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\beta}\hat{\boldsymbol{b}}_2 \right) \right) \\
&= \left( \boldsymbol{E}(\boldsymbol{\theta}) \frac{\partial \dot{\boldsymbol{\theta}}}{\partial \dot{\boldsymbol{\theta}}} \right)^\top \boldsymbol{I}_{A/C}(\beta) \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\beta}\hat{\boldsymbol{b}}_2 \right) + \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\beta}\hat{\boldsymbol{b}}_2 \right)^\top \boldsymbol{I}_{A/C}(\beta) \left( \boldsymbol{E}(\boldsymbol{\theta}) \frac{\partial \dot{\boldsymbol{\theta}}}{\partial \dot{\boldsymbol{\theta}}} \right) \\
&= 2 \left( \boldsymbol{E}(\boldsymbol{\theta}) \frac{\partial \dot{\boldsymbol{\theta}}}{\partial \dot{\boldsymbol{\theta}}} \right)^\top \boldsymbol{I}_{A/C}(\beta) \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\beta}\hat{\boldsymbol{b}}_2 \right) \\
&= 2\boldsymbol{E}(\boldsymbol{\theta})^\top \boldsymbol{I}_{A/C}(\beta) \left( \boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2 \right)
\end{aligned} \tag{39}$$

For the wheel term:

$$\begin{aligned}
&= \frac{\partial}{\partial \dot{\boldsymbol{\theta}}} \left( \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}} \right)^\top \boldsymbol{J}_w \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}} \right) \right) \\
&= \left( \boldsymbol{E}(\boldsymbol{\theta}) \frac{\partial \dot{\boldsymbol{\theta}}}{\partial \dot{\boldsymbol{\theta}}} \right)^\top \boldsymbol{J}_w \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}} \right) + \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}} \right)^\top \boldsymbol{J}_w \left( \boldsymbol{E}(\boldsymbol{\theta}) \frac{\partial \dot{\boldsymbol{\theta}}}{\partial \dot{\boldsymbol{\theta}}} \right) \\
&= 2 \left( \boldsymbol{E}(\boldsymbol{\theta}) \frac{\partial \dot{\boldsymbol{\theta}}}{\partial \dot{\boldsymbol{\theta}}} \right)^\top \boldsymbol{J}_w \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}} \right) \\
&= 2\boldsymbol{E}(\boldsymbol{\theta})^\top \boldsymbol{J}_w \left( \boldsymbol{\omega} + \boldsymbol{\Omega} \right)
\end{aligned} \tag{40}$$

Once again, in the above equations we substitute $\boldsymbol{\omega}$ for $\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}$ and $\boldsymbol{\Omega}$ for $\dot{\boldsymbol{\alpha}}$ like with the $\frac{\partial L}{\partial \boldsymbol{\theta}}$ evaluation.

So, if we substitute into Eq. (37):

$$\begin{aligned}\frac{\partial L}{\partial \dot{\boldsymbol{\theta}}} &= \frac{1}{2}\left(2\boldsymbol{E}(\boldsymbol{\theta})^\top \boldsymbol{I}_B\boldsymbol{\omega} + 2\boldsymbol{E}(\boldsymbol{\theta})^\top \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + 2\boldsymbol{E}(\boldsymbol{\theta})^\top \boldsymbol{J}_w\left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)\right) \\ &= \boldsymbol{E}(\boldsymbol{\theta})^\top\left(\boldsymbol{I}_B\boldsymbol{\omega} + \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{J}_w\left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)\right)\end{aligned} \tag{41}$$

Then we find the time derivative $\frac{d}{dt}(\frac{\partial L}{\partial \dot{\boldsymbol{\theta}}})$:

$$\begin{aligned}\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\boldsymbol{\theta}}}\right) &= \frac{d}{dt}\left(\boldsymbol{E}(\boldsymbol{\theta})^\top\left(\boldsymbol{I}_B\boldsymbol{\omega} + \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{J}_w\left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)\right)\right) \\ &= \left(\frac{d\boldsymbol{E}(\boldsymbol{\theta})}{dt}\right)^\top\left(\boldsymbol{I}_B\boldsymbol{\omega} + \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{J}_w\left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)\right) \\ &\quad + \boldsymbol{E}(\boldsymbol{\theta})^\top\frac{d}{dt}\left(\boldsymbol{I}_B\boldsymbol{\omega} + \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{J}_w\left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)\right)\end{aligned} \tag{42}$$

Then we evaluate the second term:

$$\begin{aligned}&= \frac{d}{dt}\left(\boldsymbol{I}_B\boldsymbol{\omega} + \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{J}_w\left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)\right) \\ &= \boldsymbol{I}_B\frac{d\boldsymbol{\omega}}{dt} + \frac{d\boldsymbol{I}_{A/C}(\beta)}{dt}\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{I}_{A/C}(\beta)\left(\frac{d\boldsymbol{\omega}}{dt} + \frac{d\dot{\beta}}{dt}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{J}_w\left(\frac{d\boldsymbol{\omega}}{dt} + \frac{d\boldsymbol{\Omega}}{dt}\right) \\ &= \boldsymbol{I}_B\dot{\boldsymbol{\omega}} + \dot{\boldsymbol{I}}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{I}_{A/C}(\beta)\left(\dot{\boldsymbol{\omega}} + \ddot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{J}_w\left(\dot{\boldsymbol{\omega}} + \dot{\boldsymbol{\Omega}}\right) \\ &= \boldsymbol{I}_B\dot{\boldsymbol{\omega}} + \dot{\boldsymbol{I}}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{I}_{A/C}(\beta)\dot{\boldsymbol{\omega}} + \boldsymbol{I}_{A/C}(\beta)\ddot{\beta}\hat{\boldsymbol{b}}_2 + \boldsymbol{J}_w\dot{\boldsymbol{\omega}} + \boldsymbol{J}_w\dot{\boldsymbol{\Omega}} \\ &= \left(\boldsymbol{I}_B + \boldsymbol{I}_{A/C}(\beta) + \boldsymbol{J}_w\right)\dot{\boldsymbol{\omega}} + \dot{\boldsymbol{I}}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{I}_{A/C}(\beta)\ddot{\beta}\hat{\boldsymbol{b}}_2 + \boldsymbol{J}_w\dot{\boldsymbol{\Omega}} \\ &= \boldsymbol{I}_S\dot{\boldsymbol{\omega}} + \dot{\boldsymbol{I}}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{I}_{A/C}(\beta)\ddot{\beta}\hat{\boldsymbol{b}}_2 + \boldsymbol{J}_w\dot{\boldsymbol{\Omega}} \\ &= \boldsymbol{I}_S\dot{\boldsymbol{\omega}} + \dot{\boldsymbol{I}}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{I}_{A/C}(\beta)\ddot{\beta}\hat{\boldsymbol{b}}_2 + \boldsymbol{J}_w\ddot{\boldsymbol{\alpha}}\end{aligned} \tag{43}$$

Which the predefined term $\boldsymbol{I}_S$ was used here for brevity and $\dot{\boldsymbol{\Omega}} = \ddot{\boldsymbol{\alpha}}$ since $\boldsymbol{\Omega} = \dot{\boldsymbol{\alpha}}$.

Then, we can determine the value of $\dot{\boldsymbol{\omega}}$ as:

$$\dot{\boldsymbol{\omega}} = \frac{d}{dt}\left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}\right) = \frac{d\boldsymbol{E}(\boldsymbol{\theta})}{dt}\dot{\boldsymbol{\theta}} + \boldsymbol{E}(\boldsymbol{\theta})\frac{d\dot{\boldsymbol{\theta}}}{dt} = \dot{\boldsymbol{E}}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \boldsymbol{E}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} \tag{44}$$

Then we determine the time derivative of $\boldsymbol{E}(\boldsymbol{\theta})$:

$$\begin{aligned}
\dot{\boldsymbol{E}}(\boldsymbol{\theta}) &= \frac{d}{dt}\begin{bmatrix} \sin\theta_3 \sin\theta_2 & \cos\theta_3 & 0 \\ \sin\theta_2 \cos\theta_3 & -\sin\theta_3 & 0 \\ \cos\theta_2 & 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} \frac{d}{dt}\left(\sin\theta_3 \sin\theta_2\right) & \frac{d}{dt}\left(\cos\theta_3\right) & \frac{d}{dt}\left(0\right) \\ \frac{d}{dt}\left(\sin\theta_2 \cos\theta_3\right) & \frac{d}{dt}\left(-\sin\theta_3\right) & \frac{d}{dt}\left(0\right) \\ \frac{d}{dt}\left(\cos\theta_2\right) & \frac{d}{dt}\left(0\right) & \frac{d}{dt}\left(1\right) \end{bmatrix} \\
&= \begin{bmatrix} \frac{d(\sin\theta_3)}{d\theta_3}\frac{d\theta_3}{dt}\sin\theta_2 + \sin\theta_3 \frac{d(\sin\theta_2)}{d\theta_2}\frac{d\theta_2}{dt} & \frac{d(\cos\theta_3)}{d\theta_3}\frac{d\theta_3}{dt} & 0 \\ \frac{d(\sin\theta_2)}{d\theta_2}\frac{d\theta_2}{dt}\cos\theta_3 + \sin\theta_2 \frac{d(\cos\theta_3)}{d\theta_3}\frac{d\theta_3}{dt} & -\frac{d(\sin\theta_3)}{d\theta_3}\frac{d\theta_3}{dt} & 0 \\ \frac{d(\cos\theta_2)}{d\theta_2}\frac{d\theta_2}{dt} & 0 & 0 \end{bmatrix} \\
&= \begin{bmatrix} \cos\theta_3 \sin\theta_2\, \dot{\theta}_3 + \sin\theta_3 \cos\theta_2\, \dot{\theta}_2 & -\sin\theta_3\, \dot{\theta}_3 & 0 \\ \cos\theta_2 \cos\theta_3\, \dot{\theta}_2 - \sin\theta_2 \sin\theta_3\, \dot{\theta}_3 & -\cos\theta_3\, \dot{\theta}_3 & 0 \\ -\sin\theta_2\, \dot{\theta}_2 & 0 & 0 \end{bmatrix}
\end{aligned} \tag{45}$$

Substituting the second term and $\dot{\boldsymbol{\omega}}$ back into the time derivative:

$$\begin{aligned}
&= \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\boldsymbol{\theta}}}\right) \\
&= \dot{\boldsymbol{E}}(\boldsymbol{\theta})^\top \left(\boldsymbol{I}_B \boldsymbol{\omega} + \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{J}_w\left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)\right) \\
&\quad + \boldsymbol{E}(\boldsymbol{\theta})^\top \left(\boldsymbol{I}_S \dot{\boldsymbol{\omega}} + \dot{\boldsymbol{I}}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{I}_{A/C}(\beta)\ddot{\beta}\hat{\boldsymbol{b}}_2 + \boldsymbol{J}_w \ddot{\boldsymbol{\alpha}}\right) \\
&= \dot{\boldsymbol{E}}(\boldsymbol{\theta})^\top \boldsymbol{H}_S + \boldsymbol{E}(\boldsymbol{\theta})^\top \left(\boldsymbol{I}_S \dot{\boldsymbol{\omega}} + \dot{\boldsymbol{I}}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{I}_{A/C}(\beta)\ddot{\beta}\hat{\boldsymbol{b}}_2 + \boldsymbol{J}_w \ddot{\boldsymbol{\alpha}}\right) \\
&= \dot{\boldsymbol{E}}(\boldsymbol{\theta})^\top \boldsymbol{H}_S + \boldsymbol{E}(\boldsymbol{\theta})^\top \Big(\boldsymbol{I}_S \left(\dot{\boldsymbol{E}}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \boldsymbol{E}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}}\right) + \dot{\boldsymbol{I}}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) \\
&\qquad + \boldsymbol{I}_{A/C}(\beta)\ddot{\beta}\hat{\boldsymbol{b}}_2 + \boldsymbol{J}_w \ddot{\boldsymbol{\alpha}}\Big)
\end{aligned} \tag{46}$$

Where the first parenthesis substitutes for the earlier defined $\boldsymbol{H}_S$.

Then we find the partials with respect to $\boldsymbol{\alpha}$. The Lagrangian is not a function of $\boldsymbol{\alpha}$, so the $\boldsymbol{\alpha}$ angles are all cyclic coordinates and this:

$$\frac{\partial L}{\partial \boldsymbol{\alpha}} = 0 \tag{47}$$

However, $L$ is a function of $\dot{\boldsymbol{\alpha}}$, so we find $\frac{\partial L}{\partial \dot{\boldsymbol{\alpha}}}$ as:

$$\begin{aligned}
\frac{\partial L}{\partial \dot{\boldsymbol{\alpha}}} &= \frac{1}{2}\Bigg( \frac{\partial}{\partial \dot{\boldsymbol{\alpha}}} \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} \right)^\top \boldsymbol{I}_B \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} \right) \\
&\quad + \frac{\partial}{\partial \dot{\boldsymbol{\alpha}}} \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\beta}\hat{\boldsymbol{b}}_2 \right)^\top \boldsymbol{I}_{A/C}(\beta) \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\beta}\hat{\boldsymbol{b}}_2 \right) \\
&\quad + \frac{\partial}{\partial \dot{\boldsymbol{\alpha}}} \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}} \right)^\top \boldsymbol{J}_w \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}} \right) \Bigg) \\
&= \frac{1}{2}\left( \mathbf{0} + \mathbf{0} + \frac{\partial}{\partial \dot{\boldsymbol{\alpha}}} \left( \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}} \right)^\top \boldsymbol{J}_w \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}} \right) \right) \right) \\
&= \frac{1}{2}\left( \left( \frac{\partial \dot{\boldsymbol{\alpha}}}{\partial \dot{\boldsymbol{\alpha}}} \right)^\top \boldsymbol{J}_w \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}} \right) + \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}} \right)^\top \boldsymbol{J}_w \left( \frac{\partial \dot{\boldsymbol{\alpha}}}{\partial \dot{\boldsymbol{\alpha}}} \right) \right) \\
&= \frac{1}{2}\left( 2 \left( \frac{\partial \dot{\boldsymbol{\alpha}}}{\partial \dot{\boldsymbol{\alpha}}} \right)^\top \boldsymbol{J}_w \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}} \right) \right) \\
&= \frac{1}{2}\left( 2\boldsymbol{J}_w \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}} \right) \right) \\
&= \boldsymbol{J}_w \left( \boldsymbol{\omega} + \boldsymbol{\Omega} \right)
\end{aligned} \tag{48}$$

Since there is a non-conservative force tied to $\boldsymbol{\alpha}$, then $\frac{d}{dt}(\frac{\partial L}{\partial \dot{\boldsymbol{\alpha}}})$ must be non-zero:

$$\begin{aligned}
\frac{d}{dt}\left( \frac{\partial L}{\partial \dot{\boldsymbol{\alpha}}} \right) &= \frac{d}{dt}\left( \boldsymbol{J}_w \left( \boldsymbol{\omega} + \boldsymbol{\Omega} \right) \right) \\
&= \boldsymbol{J}_w \left( \frac{d\boldsymbol{\omega}}{dt} + \frac{d\boldsymbol{\Omega}}{dt} \right) \\
&= \boldsymbol{J}_w \left( \dot{\boldsymbol{\omega}} + \dot{\boldsymbol{\Omega}} \right) \\
&= \boldsymbol{J}_w \left( \dot{\boldsymbol{E}}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \boldsymbol{E}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + \ddot{\boldsymbol{\alpha}} \right)
\end{aligned} \tag{49}$$

where $\dot{\boldsymbol{\omega}} = \dot{\boldsymbol{E}}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \boldsymbol{E}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}}$ was substituted like before and $\dot{\boldsymbol{\Omega}} = \ddot{\boldsymbol{\alpha}}$ since $\boldsymbol{\Omega} = \dot{\boldsymbol{\alpha}}$.

Then, we plug the $\boldsymbol{\theta}$ partial terms back into Lagrange's equation:

$$\begin{aligned}
\boldsymbol{Q}_\theta &= \frac{d}{dt}\left( \frac{\partial L}{\partial \dot{\boldsymbol{\theta}}} \right) - \frac{\partial L}{\partial \boldsymbol{\theta}} \\
\mathbf{0} &= \dot{\boldsymbol{E}}^\top \boldsymbol{H}_S + \boldsymbol{E}^\top \Big( \boldsymbol{I}_S \left( \dot{\boldsymbol{E}}\dot{\boldsymbol{\theta}} + \boldsymbol{E}\ddot{\boldsymbol{\theta}} \right) + \dot{\boldsymbol{I}}_{A/C}(\beta) \left( \boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2 \right) \\
&\quad + \boldsymbol{I}_{A/C}(\beta)\ddot{\beta}\hat{\boldsymbol{b}}_2 + \boldsymbol{J}_w\ddot{\boldsymbol{\alpha}} \Big) - \left( \frac{\partial \boldsymbol{E}}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}} \right)^\top \boldsymbol{H}_S \\
&= \dot{\boldsymbol{E}}^\top \boldsymbol{H}_S + \boldsymbol{E}^\top \boldsymbol{I}_S \dot{\boldsymbol{E}}\dot{\boldsymbol{\theta}} + \boldsymbol{E}^\top \boldsymbol{I}_S \boldsymbol{E}\ddot{\boldsymbol{\theta}} + \boldsymbol{E}^\top \dot{\boldsymbol{I}}_{A/C}(\beta) \left( \boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2 \right) \\
&\quad + \boldsymbol{E}^\top \boldsymbol{I}_{A/C}(\beta)\ddot{\beta}\hat{\boldsymbol{b}}_2 + \boldsymbol{E}^\top \boldsymbol{J}_w\ddot{\boldsymbol{\alpha}} - \left( \frac{\partial \boldsymbol{E}}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}} \right)^\top \boldsymbol{H}_S
\end{aligned} \tag{50}$$

And isolate the $\ddot{\boldsymbol{q}}$ terms on the left side:

$$\begin{aligned}\boldsymbol{E}^\top \boldsymbol{I}_S \boldsymbol{E}\ddot{\boldsymbol{\theta}} + \boldsymbol{E}^\top \boldsymbol{J}_w \ddot{\boldsymbol{\alpha}} = &-\dot{\boldsymbol{E}}^\top \boldsymbol{H}_S - \boldsymbol{E}^\top \boldsymbol{I}_S \dot{\boldsymbol{E}}\dot{\boldsymbol{\theta}} - \boldsymbol{E}^\top \dot{\boldsymbol{I}}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) \\ &- \boldsymbol{E}^\top \boldsymbol{I}_{A/C}(\beta)\ddot{\beta}\hat{\boldsymbol{b}}_2 + \left(\frac{\partial \boldsymbol{E}}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}}\right)^\top \boldsymbol{H}_S\end{aligned} \tag{51}$$

Then we do the same for the $\boldsymbol{\alpha}$ partial terms for Lagrange's equation:

$$\begin{aligned}\boldsymbol{Q}_\alpha &= \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\boldsymbol{\alpha}}}\right) - \frac{\partial L}{\partial \boldsymbol{\alpha}} \\ \boldsymbol{u} &= \boldsymbol{J}_w\left(\dot{\boldsymbol{E}}\dot{\boldsymbol{\theta}} + \boldsymbol{E}\ddot{\boldsymbol{\theta}} + \ddot{\boldsymbol{\alpha}}\right) - \boldsymbol{0} \\ &= \boldsymbol{J}_w \dot{\boldsymbol{E}}\dot{\boldsymbol{\theta}} + \boldsymbol{J}_w \boldsymbol{E}\ddot{\boldsymbol{\theta}} + \boldsymbol{J}_w \ddot{\boldsymbol{\alpha}}\end{aligned} \tag{52}$$

And isolate the $\ddot{\boldsymbol{q}}$ terms on the left side:

$$\boldsymbol{J}_w \boldsymbol{E}\ddot{\boldsymbol{\theta}} + \boldsymbol{J}_w \ddot{\boldsymbol{\alpha}} = \boldsymbol{u} - \boldsymbol{J}_w \dot{\boldsymbol{E}}\dot{\boldsymbol{\theta}} \tag{53}$$

Then we can take the left sides of Eq. (51) and Eq. (53) to make the $\boldsymbol{M}$ matrix requested by the problem:

$$\boldsymbol{M}(\boldsymbol{q}, t)\ddot{\boldsymbol{q}} = \begin{bmatrix} \boldsymbol{E}^\top \boldsymbol{I}_S \boldsymbol{E} & \boldsymbol{E}^\top \boldsymbol{J}_w \\ \boldsymbol{J}_w \boldsymbol{E} & \boldsymbol{J}_w \end{bmatrix} \begin{bmatrix} \ddot{\boldsymbol{\theta}} \\ \ddot{\boldsymbol{\alpha}} \end{bmatrix} \tag{54}$$

And the right sides of Eq. (51) and Eq. (53) make the following system of equations $\boldsymbol{f}(\boldsymbol{q}, \dot{\boldsymbol{q}}, t, u)$:

$$\begin{aligned}\boldsymbol{f} &= \begin{bmatrix} -\dot{\boldsymbol{E}}^\top \boldsymbol{H}_S - \boldsymbol{E}^\top \boldsymbol{I}_S \dot{\boldsymbol{E}}\dot{\boldsymbol{\theta}} - \boldsymbol{E}^\top \dot{\boldsymbol{I}}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) - \boldsymbol{E}^\top \boldsymbol{I}_{A/C}(\beta)\ddot{\beta}\hat{\boldsymbol{b}}_2 + \left(\frac{\partial \boldsymbol{E}}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}}\right)^\top \boldsymbol{H}_S \\ \boldsymbol{u} - \boldsymbol{J}_w \dot{\boldsymbol{E}}\dot{\boldsymbol{\theta}} \end{bmatrix} \\ &= \begin{bmatrix} -\dot{\boldsymbol{E}}^\top \boldsymbol{H}_S - \boldsymbol{E}^\top \left(\boldsymbol{I}_S \dot{\boldsymbol{E}}\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{I}}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{I}_{A/C}(\beta)\ddot{\beta}\hat{\boldsymbol{b}}_2\right) + \left(\frac{\partial \boldsymbol{E}}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}}\right)^\top \boldsymbol{H}_S \\ \boldsymbol{u} - \boldsymbol{J}_w \dot{\boldsymbol{E}}\dot{\boldsymbol{\theta}} \end{bmatrix} \\ &= \begin{bmatrix} -\boldsymbol{E}^\top \left(\boldsymbol{I}_S \dot{\boldsymbol{E}}\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{I}}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{I}_{A/C}(\beta)\ddot{\beta}\hat{\boldsymbol{b}}_2\right) + \left(\frac{\partial \boldsymbol{E}}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}} - \dot{\boldsymbol{E}}\right)^\top \boldsymbol{H}_S \\ \boldsymbol{u} - \boldsymbol{J}_w \dot{\boldsymbol{E}}\dot{\boldsymbol{\theta}} \end{bmatrix}\end{aligned} \tag{55}$$

We see in some of the equations derived in this section, there are first and second derivative terms of the $\beta$ angle. We can derive those terms as follows.

In Eq. (3), we see that $\tau$ is directly a function of $t$ divided by a constant, thus the function for $\dot{\beta}$ can be found easily:

$$\begin{aligned}\dot{\beta}(t) &= \begin{cases} \beta_m \left(10\frac{d}{dt}\left(\frac{t}{t_m}\right)^3 - 15\frac{d}{dt}\left(\frac{t}{t_m}\right)^4 + 6\frac{d}{dt}\left(\frac{t}{t_m}\right)^5\right), & 0 \le t \le t_m, \\ \frac{d}{dt}\left(\beta_m\right), & t > t_m, \end{cases} \\ &= \begin{cases} \beta_m \left(\frac{30}{t_m}\left(\frac{t}{t_m}\right)^2 - \frac{60}{t_m}\left(\frac{t}{t_m}\right)^3 + \frac{30}{t_m}\left(\frac{t}{t_m}\right)^4\right), & 0 \le t \le t_m, \\ 0, & t > t_m, \end{cases} \\ &= \begin{cases} \frac{\beta_m}{t_m}\left(30\tau^2 - 60\tau^3 + 30\tau^4\right), & 0 \le t \le t_m, \\ 0, & t > t_m, \end{cases}\end{aligned} \tag{56}$$

Then we perform the next derivative:

$$\begin{aligned}
\ddot{\beta}(t) &= \begin{cases} \frac{\beta_m}{t_m}\left(30\frac{d}{dt}\left(\frac{t}{t_m}\right)^2 - 60\frac{d}{dt}\left(\frac{t}{t_m}\right)^3 + 30\frac{d}{dt}\left(\frac{t}{t_m}\right)^4\right), & 0 \le t \le t_m, \\ \frac{d}{dt}(0), & t > t_m, \end{cases} \\
&= \begin{cases} \frac{\beta_m}{t_m}\left(\frac{60}{t_m}\left(\frac{t}{t_m}\right) - \frac{180}{t_m}\left(\frac{t}{t_m}\right)^2 + \frac{120}{t_m}\left(\frac{t}{t_m}\right)^3\right), & 0 \le t \le t_m, \\ 0, & t > t_m, \end{cases} \\
&= \begin{cases} \frac{\beta_m}{t_m^2}\left(60\tau - 180\tau^2 + 120\tau^3\right), & 0 \le t \le t_m, \\ 0, & t > t_m, \end{cases}
\end{aligned} \tag{57}$$

Then, recall for the given function for the $\boldsymbol{I}_A(\beta)$ matrix, we need the rotation matrix $\boldsymbol{R}_2(\beta)$ which is simply a second-axis rotation in the body frame since all above derived equations maintain body frame components. As seen in the notation, this rotation matrix is a function of the $\beta$ angle representing the current orientation of the solar array's rotation:

$$\boldsymbol{R}_2(\beta) = \begin{bmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{bmatrix} \tag{58}$$

So, below the functions for $\beta$ and its derivatives used in the EoM's are restated for reporting:

$$\begin{aligned}
\beta(t) &= \begin{cases} \beta_m\left(10\tau^3 - 15\tau^4 + 6\tau^5\right), & 0 \le t \le t_m, \\ \beta_m, & t > t_m, \end{cases} \qquad \tau \coloneqq \frac{t}{t_m} \\
\dot{\beta}(t) &= \begin{cases} \frac{\beta_m}{t_m}\left(30\tau^2 - 60\tau^3 + 30\tau^4\right), & 0 \le t \le t_m, \\ 0, & t > t_m, \end{cases} \\
\ddot{\beta}(t) &= \begin{cases} \frac{\beta_m}{t_m^2}\left(60\tau - 180\tau^2 + 120\tau^3\right), & 0 \le t \le t_m, \\ 0, & t > t_m, \end{cases}
\end{aligned} \tag{59}$$

And the the final EoM are represented as:

$$\boldsymbol{M}(\boldsymbol{q},t)\ddot{\boldsymbol{q}} = \boldsymbol{f}(\boldsymbol{q},\dot{\boldsymbol{q}},t,u) \tag{60}$$

Where:

$$\boldsymbol{q} = \begin{bmatrix}\boldsymbol{\theta}\\ \boldsymbol{\alpha}\end{bmatrix}, \qquad \dot{\boldsymbol{q}} = \begin{bmatrix}\dot{\boldsymbol{\theta}}\\ \dot{\boldsymbol{\alpha}}\end{bmatrix}, \qquad \ddot{\boldsymbol{q}} = \begin{bmatrix}\ddot{\boldsymbol{\theta}}\\ \ddot{\boldsymbol{\alpha}}\end{bmatrix}$$

$$\boldsymbol{M}(\boldsymbol{q},t) = \begin{bmatrix}\boldsymbol{E}^\top \boldsymbol{I}_S \boldsymbol{E} & \boldsymbol{E}^\top \boldsymbol{J}_w\\ \boldsymbol{J}_w \boldsymbol{E} & \boldsymbol{J}_w\end{bmatrix}$$

$$\boldsymbol{f}(\boldsymbol{q},\dot{\boldsymbol{q}},t,u) = \begin{bmatrix}-\boldsymbol{E}^\top \left(\boldsymbol{I}_S \dot{\boldsymbol{E}}\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{I}}_{A/C}(\boldsymbol{E}\dot{\boldsymbol{\theta}} + \dot{\beta}\hat{\boldsymbol{b}}_2) + \boldsymbol{I}_{A/C}\ddot{\beta}\hat{\boldsymbol{b}}_2\right) + \left(\frac{\partial \boldsymbol{E}}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}} - \dot{\boldsymbol{E}}\right)^\top \boldsymbol{H}_S\\ \boldsymbol{u} - \boldsymbol{J}_w \dot{\boldsymbol{E}}\dot{\boldsymbol{\theta}}\end{bmatrix}$$

$$\boldsymbol{H}_S = \boldsymbol{I}_B \boldsymbol{E}\dot{\boldsymbol{\theta}} + \boldsymbol{I}_{A/C}\left(\boldsymbol{E}\dot{\boldsymbol{\theta}} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{J}_w\left(\boldsymbol{E}\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}}\right)$$

$$\boldsymbol{I}_S = \boldsymbol{I}_B + \boldsymbol{I}_{A/C} + \boldsymbol{J}_w$$

$$\boldsymbol{I}_{A/C}(\beta) = 2\boldsymbol{I}_A(\beta) + 2m_A d^2\left(\boldsymbol{I}_3 - \hat{\boldsymbol{b}}_2\hat{\boldsymbol{b}}_2^\top\right)$$

$$\boldsymbol{I}_A(\beta) = \boldsymbol{R}_2(\beta)\boldsymbol{I}_{A0}\boldsymbol{R}_2^\top(\beta)$$

$$\boldsymbol{R}_2(\beta) = \begin{bmatrix}\cos\beta & 0 & \sin\beta\\ 0 & 1 & 0\\ -\sin\beta & 0 & \cos\beta\end{bmatrix}$$

$$\boldsymbol{E}(\boldsymbol{\theta}) = \begin{bmatrix}\sin\theta_3\sin\theta_2 & \cos\theta_3 & 0\\ \sin\theta_2\cos\theta_3 & -\sin\theta_3 & 0\\ \cos\theta_2 & 0 & 1\end{bmatrix}$$

$$\dot{\boldsymbol{E}}(\boldsymbol{\theta},\dot{\boldsymbol{\theta}}) = \begin{bmatrix}\cos\theta_3\sin\theta_2\,\dot{\theta}_3 + \sin\theta_3\cos\theta_2\,\dot{\theta}_2 & -\sin\theta_3\,\dot{\theta}_3 & 0\\ \cos\theta_2\cos\theta_3\,\dot{\theta}_2 - \sin\theta_2\sin\theta_3\,\dot{\theta}_3 & -\cos\theta_3\,\dot{\theta}_3 & 0\\ -\sin\theta_2\,\dot{\theta}_2 & 0 & 0\end{bmatrix}$$

$$\frac{\partial \boldsymbol{E}}{\partial \boldsymbol{\theta}}\dot{\boldsymbol{\theta}} = \begin{bmatrix}\frac{\partial \boldsymbol{E}}{\partial \theta_1}\dot{\boldsymbol{\theta}} & \frac{\partial \boldsymbol{E}}{\partial \theta_2}\dot{\boldsymbol{\theta}} & \frac{\partial \boldsymbol{E}}{\partial \theta_3}\dot{\boldsymbol{\theta}}\end{bmatrix}, \qquad \frac{\partial \boldsymbol{E}}{\partial \boldsymbol{\theta}} = \left\{\frac{\partial \boldsymbol{E}}{\partial \theta_1}, \frac{\partial \boldsymbol{E}}{\partial \theta_2}, \frac{\partial \boldsymbol{E}}{\partial \theta_3}\right\}$$

$$\frac{\partial \boldsymbol{E}}{\partial \theta_1} = \boldsymbol{0}_{3\times 3}$$

$$\frac{\partial \boldsymbol{E}}{\partial \theta_2} = \begin{bmatrix}\sin\theta_3\cos\theta_2 & 0 & 0\\ \cos\theta_2\cos\theta_3 & 0 & 0\\ -\sin\theta_2 & 0 & 0\end{bmatrix}$$

$$\frac{\partial \boldsymbol{E}}{\partial \theta_3} = \begin{bmatrix}\cos\theta_3\sin\theta_2 & -\sin\theta_3 & 0\\ -\sin\theta_2\sin\theta_3 & -\cos\theta_3 & 0\\ 0 & 0 & 0\end{bmatrix}$$

(61)

**(a.5): Find**

Show that $\boldsymbol{\alpha} \in \mathbb{R}^3$ is cyclic, and derive its generalized momentum, denoted by $\boldsymbol{p}_\alpha \in \mathcal{R}^3$.

**Analysis/Conclusions:**

Its is known that a set of coordinates is cyclic if there is continuous symmetry, which can be determined by saying the Lagrangian is not a function of those coordinates. We saw in (a.4) that the Lagrangian:

$$\begin{aligned} L = \frac{1}{2}\Bigg( & \left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}\right)^\top \boldsymbol{I}_B \left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}\right) \\ & + \left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\beta}\hat{\boldsymbol{b}}_2\right)^\top \boldsymbol{I}_{A/C}(\beta) \left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) \\ & + \left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}}\right)^\top \boldsymbol{J}_w \left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}}\right) \Bigg) \end{aligned} \tag{62}$$

$$\text{Where:} \quad \boldsymbol{E}(\boldsymbol{\theta}) = \begin{bmatrix} \sin\theta_3 \sin\theta_2 & \cos\theta_3 & 0 \\ \sin\theta_2 \cos\theta_3 & -\sin\theta_3 & 0 \\ \cos\theta_2 & 0 & 1 \end{bmatrix}$$

is not a function of $\boldsymbol{\alpha}$, which means $\boldsymbol{\alpha}$ **is cyclic**.

The generalized momentum term $\boldsymbol{p}_\alpha$ is then given by the term from the Lagrangian $\frac{\partial L}{\partial \dot{\boldsymbol{\alpha}}}$ which was also found in (a.4) as:

$$\boldsymbol{p}_\alpha = \frac{\partial L}{\partial \dot{\boldsymbol{\alpha}}} = \boldsymbol{J}_w \left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}}\right) = \boldsymbol{J}_w \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) \tag{63}$$

**(a.6): Find**

Derive the time derivative of $\boldsymbol{p_\alpha}$, and discuss when it is conserved.

**Analysis/Conclusions:**

The time derivative $\frac{d}{dt}\boldsymbol{p_\alpha}$ is equal to the time derivative $\frac{d}{dt}(\frac{\partial L}{\partial \dot{\boldsymbol{\alpha}}})$ which was also found in (a.4) as part of the Lagrange equation for $\boldsymbol{\alpha}$:

$$\dot{\boldsymbol{p}}_{\boldsymbol{\alpha}} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\boldsymbol{\alpha}}}\right) = \boldsymbol{J}_w\left(\dot{\boldsymbol{E}}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \boldsymbol{E}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + \ddot{\boldsymbol{\alpha}}\right) \tag{64}$$

Since $\boldsymbol{\alpha}$ is cyclic, we see from the Lagrange equation that the $\frac{\partial L}{\partial \boldsymbol{\alpha}}$ term equals zero making the time derivative of generalized momentum $\boldsymbol{p_\alpha}$ equal the motor torques $\boldsymbol{u}$:

$$\begin{aligned} \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\boldsymbol{\alpha}}}\right) - \frac{\partial L}{\partial \boldsymbol{\alpha}} &= \boldsymbol{Q}_\alpha \\ \dot{\boldsymbol{p}}_{\boldsymbol{\alpha}} - \boldsymbol{0} &= \boldsymbol{u} \\ \dot{\boldsymbol{p}}_{\boldsymbol{\alpha}} &= \boldsymbol{u} \end{aligned} \tag{65}$$

This means that the generalized momentum $\boldsymbol{p_\alpha}$ is **only conserved when the motors are not running and thus not providing torque**.

**(a.7): Find**

Calculate the Jacobi integral $\mathcal{J}$ and its time derivative, and discuss if it is conserved.

**Analysis/Conclusions:**

We can find the Jacobi integral through:

$$\mathcal{J} = \dot{\boldsymbol{q}}^\top \frac{\partial L}{\partial \dot{\boldsymbol{q}}} - L = \dot{\boldsymbol{\theta}}^\top \frac{\partial L}{\partial \dot{\boldsymbol{\theta}}} + \dot{\boldsymbol{\alpha}}^\top \frac{\partial L}{\partial \dot{\boldsymbol{\alpha}}} - L \tag{66}$$

So first we find the term $\dot{\boldsymbol{\theta}}^\top \frac{\partial L}{\partial \dot{\boldsymbol{\theta}}}$ which has been partially found from terms in (a.4):

$$\begin{aligned} \dot{\boldsymbol{\theta}}^\top \frac{\partial L}{\partial \dot{\boldsymbol{\theta}}} &= \dot{\boldsymbol{\theta}}^\top \boldsymbol{E}(\boldsymbol{\theta})^\top \left(\boldsymbol{I}_B\boldsymbol{\omega} + \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{J}_w\left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)\right) \\ &= \left(\boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}\right)^\top \left(\boldsymbol{I}_B\boldsymbol{\omega} + \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{J}_w\left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)\right) \\ &= \boldsymbol{\omega}^\top \boldsymbol{I}_B\boldsymbol{\omega} + \boldsymbol{\omega}^\top \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{\omega}^\top \boldsymbol{J}_w\left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) \end{aligned} \tag{67}$$

Likewise, so were the terms of $\dot{\boldsymbol{\alpha}}^\top \frac{\partial L}{\partial \dot{\boldsymbol{\alpha}}}$:

$$\dot{\boldsymbol{\alpha}}^\top \frac{\partial L}{\partial \dot{\boldsymbol{\alpha}}} = \dot{\boldsymbol{\alpha}}^\top \boldsymbol{J}_w \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) = \boldsymbol{\Omega}^\top \boldsymbol{J}_w \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) \tag{68}$$

And then we combine these terms back into Eq (66) using the Lagrangian $L$ found in (a.3):

$$\begin{aligned}
\mathcal{J} &= \dot{\boldsymbol{\theta}}^\top \frac{\partial L}{\partial \dot{\boldsymbol{\theta}}} + \dot{\boldsymbol{\alpha}}^\top \frac{\partial L}{\partial \dot{\boldsymbol{\alpha}}} - L \\
&= \boldsymbol{\omega}^\top \boldsymbol{I}_B \boldsymbol{\omega} + \boldsymbol{\omega}^\top \boldsymbol{I}_{A/C}(\beta) \left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{\omega}^\top \boldsymbol{J}_w \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) + \boldsymbol{\Omega}^\top \boldsymbol{J}_w \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) \\
&\quad - \frac{1}{2}\left(\boldsymbol{\omega}^\top \boldsymbol{I}_B \boldsymbol{\omega} + \left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right)^\top \boldsymbol{I}_{A/C}(\beta) \left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)^\top \boldsymbol{J}_w \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)\right)
\end{aligned} \tag{69}$$

Some of those terms in the Lagrangian multiply out to make:

$$\begin{aligned}
&= \boldsymbol{\omega}^\top \boldsymbol{I}_{A/C}(\beta) \left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) \\
&= \boldsymbol{\omega}^\top \boldsymbol{I}_{A/C}(\beta)\boldsymbol{\omega} + \dot{\beta}\, \boldsymbol{\omega}^\top \boldsymbol{I}_{A/C}(\beta)\hat{\boldsymbol{b}}_2
\end{aligned}$$

$$\begin{aligned}
&= \left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right)^\top \boldsymbol{I}_{A/C}(\beta) \left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) \\
&= \boldsymbol{\omega}^\top \boldsymbol{I}_{A/C}(\beta)\boldsymbol{\omega} + \dot{\beta}\, \boldsymbol{\omega}^\top \boldsymbol{I}_{A/C}(\beta)\hat{\boldsymbol{b}}_2 + \dot{\beta}\, \hat{\boldsymbol{b}}_2^\top \boldsymbol{I}_{A/C}(\beta)\boldsymbol{\omega} + \dot{\beta}^2 \hat{\boldsymbol{b}}_2^\top \boldsymbol{I}_{A/C}(\beta)\hat{\boldsymbol{b}}_2 \\
&= \boldsymbol{\omega}^\top \boldsymbol{I}_{A/C}(\beta)\boldsymbol{\omega} + 2\dot{\beta}\, \boldsymbol{\omega}^\top \boldsymbol{I}_{A/C}(\beta)\hat{\boldsymbol{b}}_2 + \dot{\beta}^2 \hat{\boldsymbol{b}}_2^\top \boldsymbol{I}_{A/C}(\beta)\hat{\boldsymbol{b}}_2
\end{aligned} \tag{70}$$

$$\begin{aligned}
&= \boldsymbol{\omega}^\top \boldsymbol{J}_w \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) + \boldsymbol{\Omega}^\top \boldsymbol{J}_w \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) \\
&= \left(\boldsymbol{\omega}^\top + \boldsymbol{\Omega}^\top\right) \boldsymbol{J}_w \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) \\
&= \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)^\top \boldsymbol{J}_w \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)
\end{aligned}$$

Which then simplifies to get the Jacobi integral:

$$\begin{aligned}
\mathcal{J} &= \boldsymbol{\omega}^\top \boldsymbol{I}_B \boldsymbol{\omega} + \boldsymbol{\omega}^\top \boldsymbol{I}_{A/C}(\beta)\boldsymbol{\omega} + \dot{\beta}\, \boldsymbol{\omega}^\top \boldsymbol{I}_{A/C}(\beta)\hat{\boldsymbol{b}}_2 + \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)^\top \boldsymbol{J}_w \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) \\
&\quad - \frac{1}{2}\boldsymbol{\omega}^\top \boldsymbol{I}_B \boldsymbol{\omega} - \frac{1}{2}\boldsymbol{\omega}^\top \boldsymbol{I}_{A/C}(\beta)\boldsymbol{\omega} - \dot{\beta}\, \boldsymbol{\omega}^\top \boldsymbol{I}_{A/C}(\beta)\hat{\boldsymbol{b}}_2 - \frac{1}{2}\dot{\beta}^2 \hat{\boldsymbol{b}}_2^\top \boldsymbol{I}_{A/C}(\beta)\hat{\boldsymbol{b}}_2 - \frac{1}{2}\left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)^\top \boldsymbol{J}_w \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) \\
&= \frac{1}{2}\boldsymbol{\omega}^\top \boldsymbol{I}_B \boldsymbol{\omega} + \frac{1}{2}\boldsymbol{\omega}^\top \boldsymbol{I}_{A/C}(\beta)\boldsymbol{\omega} + \frac{1}{2}\left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)^\top \boldsymbol{J}_w \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) - \frac{1}{2}\dot{\beta}^2 \hat{\boldsymbol{b}}_2^\top \boldsymbol{I}_{A/C}(\beta)\hat{\boldsymbol{b}}_2 \\
&= \frac{1}{2}\left(\boldsymbol{\omega}^\top \boldsymbol{I}_B \boldsymbol{\omega} + \boldsymbol{\omega}^\top \boldsymbol{I}_{A/C}(\beta)\boldsymbol{\omega} + \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right)^\top \boldsymbol{J}_w \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) - \dot{\beta}^2 \hat{\boldsymbol{b}}_2^\top \boldsymbol{I}_{A/C}(\beta)\hat{\boldsymbol{b}}_2\right)
\end{aligned} \tag{71}$$

So our Jacobi integral evaluated to be:

$$\mathcal{J} = \frac{1}{2}\left(\boldsymbol{\omega}^\top \boldsymbol{I}_B \boldsymbol{\omega} + \boldsymbol{\omega}^\top \boldsymbol{I}_{A/C}(\beta)\boldsymbol{\omega} + (\boldsymbol{\omega}+\boldsymbol{\Omega})^\top \boldsymbol{J}_w (\boldsymbol{\omega}+\boldsymbol{\Omega}) - \dot{\beta}^2 \hat{\boldsymbol{b}}_2^\top \boldsymbol{I}_{A/C}(\beta)\hat{\boldsymbol{b}}_2\right) \tag{72}$$

We then know the time derivative of the Jacobi integral can be written as:

$$\dot{\mathcal{J}} = \sum_{j=1}^{n} Q_{nc,j}\dot{q}_j - \frac{\partial L}{\partial t} \tag{73}$$

Where $Q_{nc,j}$ are the non-conservative forces.

For this system, $\boldsymbol{Q}_\theta = \boldsymbol{0}$ and $\boldsymbol{Q}_\alpha = \boldsymbol{u}$ as found in (a.4), so the non-conservative term becomes:

$$\sum_{j=1}^{n} Q_{nc,j}\dot{q}_j = \dot{\boldsymbol{\theta}}^\top \boldsymbol{Q}_\theta + \dot{\boldsymbol{\alpha}}^\top \boldsymbol{Q}_\alpha = \dot{\boldsymbol{\theta}}^\top \boldsymbol{0} + \dot{\boldsymbol{\alpha}}^\top \boldsymbol{u} = \boldsymbol{u}^\top \boldsymbol{\Omega} \tag{74}$$

Then, the Lagrangian depends explicitly on $t$ through $\beta(t)$ and $\dot{\beta}(t)$, so the time partial is non-zero:

$$\begin{aligned}
\frac{\partial L}{\partial t} &= \frac{\partial}{\partial t}\left(\frac{1}{2}\left(\boldsymbol{\omega}^\top \boldsymbol{I}_B \boldsymbol{\omega} + \left(\boldsymbol{\omega}+\dot{\beta}\hat{\boldsymbol{b}}_2\right)^\top \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{\omega}+\dot{\beta}\hat{\boldsymbol{b}}_2\right) + (\boldsymbol{\omega}+\boldsymbol{\Omega})^\top \boldsymbol{J}_w(\boldsymbol{\omega}+\boldsymbol{\Omega})\right)\right) \\
&= \frac{1}{2}\frac{\partial}{\partial t}\left(\boldsymbol{\omega}^\top \boldsymbol{I}_B \boldsymbol{\omega}\right) + \frac{1}{2}\frac{\partial}{\partial t}\left(\left(\boldsymbol{\omega}+\dot{\beta}\hat{\boldsymbol{b}}_2\right)^\top \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{\omega}+\dot{\beta}\hat{\boldsymbol{b}}_2\right)\right) + \frac{1}{2}\frac{\partial}{\partial t}\left((\boldsymbol{\omega}+\boldsymbol{\Omega})^\top \boldsymbol{J}_w(\boldsymbol{\omega}+\boldsymbol{\Omega})\right) \\
&= 0 + \frac{1}{2}\frac{\partial}{\partial t}\left(\left(\boldsymbol{\omega}+\dot{\beta}\hat{\boldsymbol{b}}_2\right)^\top \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{\omega}+\dot{\beta}\hat{\boldsymbol{b}}_2\right)\right) + 0 \\
&= \frac{1}{2}\left(\frac{\partial \boldsymbol{\omega}}{\partial t} + \frac{d\dot{\beta}}{dt}\hat{\boldsymbol{b}}_2\right)^\top \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{\omega}+\dot{\beta}\hat{\boldsymbol{b}}_2\right) + \frac{1}{2}\left(\boldsymbol{\omega}+\dot{\beta}\hat{\boldsymbol{b}}_2\right)^\top \frac{\partial \boldsymbol{I}_{A/C}(\beta)}{\partial \beta}\frac{d\beta}{dt}\left(\boldsymbol{\omega}+\dot{\beta}\hat{\boldsymbol{b}}_2\right) \\
&\quad + \frac{1}{2}\left(\boldsymbol{\omega}+\dot{\beta}\hat{\boldsymbol{b}}_2\right)^\top \boldsymbol{I}_{A/C}(\beta)\left(\frac{\partial \boldsymbol{\omega}}{\partial t} + \frac{d\dot{\beta}}{dt}\hat{\boldsymbol{b}}_2\right) \\
&= \frac{1}{2}\left(\ddot{\beta}\hat{\boldsymbol{b}}_2\right)^\top \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{\omega}+\dot{\beta}\hat{\boldsymbol{b}}_2\right) + \frac{1}{2}\left(\boldsymbol{\omega}+\dot{\beta}\hat{\boldsymbol{b}}_2\right)^\top \dot{\boldsymbol{I}}_{A/C}(\beta)\left(\boldsymbol{\omega}+\dot{\beta}\hat{\boldsymbol{b}}_2\right) \\
&\quad + \frac{1}{2}\left(\boldsymbol{\omega}+\dot{\beta}\hat{\boldsymbol{b}}_2\right)^\top \boldsymbol{I}_{A/C}(\beta)\left(\ddot{\beta}\hat{\boldsymbol{b}}_2\right) \\
&= \frac{1}{2}\left(\boldsymbol{\omega}+\dot{\beta}\hat{\boldsymbol{b}}_2\right)^\top \dot{\boldsymbol{I}}_{A/C}(\beta)\left(\boldsymbol{\omega}+\dot{\beta}\hat{\boldsymbol{b}}_2\right) + \ddot{\beta}\,\hat{\boldsymbol{b}}_2^\top \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{\omega}+\dot{\beta}\hat{\boldsymbol{b}}_2\right)
\end{aligned} \tag{75}$$

Substituting both back in gives the time derivative of the Jacobi integral:

$$\dot{\mathcal{J}} = \boldsymbol{u}^\top \boldsymbol{\Omega} - \frac{1}{2}\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right)^\top \dot{\boldsymbol{I}}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) - \ddot{\beta}\,\hat{\boldsymbol{b}}_2^\top \boldsymbol{I}_{A/C}(\beta)\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) \tag{76}$$

We know that the angle $\beta$ in the Lagrangian is prescribed explicitly by time through $\tau \coloneqq \frac{t}{t_m}$. We also know that there are non-conservative forces given by the motor torques $\boldsymbol{u}$. This means that for both these reasons $\mathcal{J}$ **is not conserved**.

However, for $t > t_m$ the arrays hold at $\beta_m$ so $\dot{\beta} = \ddot{\beta} = 0$ and $\frac{\partial L}{\partial t} = 0$, so if $\boldsymbol{u} = \boldsymbol{0}$ as well, $\mathcal{J}$ is conserved after that time if there are also no control inputs.

This makes sense since, the motor torques $\boldsymbol{u}$ and the angle $\beta$ both effectively represent (or correlate to in the case of $\beta$) torques imposed on the system by motors, which is bringing energy into the system from somewhere not considered by the generalized coordinates.

## (b): Given

Consider the attitude behavior with zero RW motor torque $\boldsymbol{u}(t) = \mathbf{0}$ with the initial spin rate $\boldsymbol{\Omega}(0) = \boldsymbol{\Omega}_0$, while the solar arrays are rotating according to Eq. (3). You can use Eq. (77) without derivation:

$$\dot{\boldsymbol{I}}_{A/C}(t) = 2\dot{\boldsymbol{I}}_A(t) = 2\dot{\beta}(t)\left(\left[\hat{\boldsymbol{b}}_2\right]_\times \boldsymbol{I}_A(t) - \boldsymbol{I}_A(t)\left[\hat{\boldsymbol{b}}_2\right]_\times\right) \tag{77}$$

For the numerical work, use the following parameters and initial conditions:

$$\begin{aligned}
&\boldsymbol{I}_B = \text{diag}(80, 100, 60)\ \text{kg}\,\text{m}^2, \quad m_A = 24\ \text{kg}, \quad d = 1.5\ \text{m},\\
&\boldsymbol{I}_{A0} = \text{diag}(6, 30, 35)\ \text{kg}\,\text{m}^2, \quad \boldsymbol{J}_w = \text{diag}(0.10, 0.12, 0.08)\ \text{kg}\,\text{m}^2,\\
&\beta_m = 60\ \text{deg}, \quad t_m = 60\ \text{s},\\
&\boldsymbol{\theta}(0) = [\pi/3, \pi/3, \pi/3]^\top\ \text{rad}, \quad \boldsymbol{\omega}(0) = \mathbf{0}, \quad \boldsymbol{\alpha}(0) = \mathbf{0},\\
&\boldsymbol{\Omega}(0) = \boldsymbol{\Omega}_0 = [300, -200, 100]^\top\ \text{rad/s}
\end{aligned} \tag{78}$$

### (b.1): Find

Plot the time history of $\beta$ and $\dot{\beta}$ based on Eq. (3).

**Analysis/Conclusions:**

In (a.4) the equations for $\beta$ and $\dot{\beta}$ were found in Eq. (59). Likewise, the results of both equations plotted from 0 seconds to 100 seconds is seen in Fig. (2):

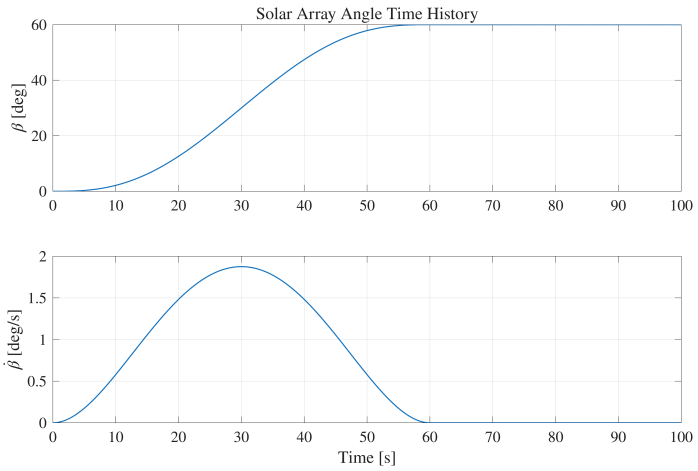



Figure 2: Time histories of the array angle $\beta$ and rate $\dot{\beta}$ over $0 \leq t \leq 100$

**(b.2): Find**

Implement the EoM derived above, and report the implemented code. Numerically propagate the system for $0 \leq t \leq 100$ sec with $\boldsymbol{u} = \boldsymbol{0}$. Plot the time histories $\boldsymbol{\theta}$, $\boldsymbol{\omega}$, and $\boldsymbol{\Omega}$.

**Analysis/Conclusions:**

The given term for $\dot{\boldsymbol{I}}_{A/C}(t)$ in the problem setup adds a skew-symmetric matrix for the body frame vector $\hat{\boldsymbol{b}}_2$ which is represented as:

$$\begin{aligned}\left[\hat{\boldsymbol{b}}_2\right]_\times &= \begin{bmatrix} 0 & -b_{2,3} & b_{2,2} \\ b_{2,3} & 0 & -b_{2,1} \\ -b_{2,2} & b_{2,1} & 0 \end{bmatrix}, \qquad \hat{\boldsymbol{b}}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}\end{aligned} \tag{79}$$

Which we are allowed to use the formulation this way because the EoM's in part (a) were derived using body frame coordinate components.

Likewise, the implementation of the EoM's between Eq. (59) Eq. (61), Eq. (77), and Eq. (79) can be seen in the listing blocks below.

First, the equations of motion were put into a function block to be ran with a numerical integrator.

Listing 1: Implementation of the spacecraft EoM.

```
function xd = rwEoM(x, t, u, bt, bdt, bddt, IB, IA0, Jw, mA, d)

   % Break the state vector into its components
   th = x(1:3); thd = x(7:9); ad = x(10:12);

   % Find the value of the beta angles for this time step
   b = bt(t); bd = bdt(t); bdd = bddt(t);

   % Define 2nd axis body frame vector and its skew-symmetric matrix
   bh2 = [0; 1; 0];
   bh2x = [0 0 1; 0 0 0; -1 0 0];

   % Define the 2nd axis rotation about b2 body frame vector
   R2 = [cos(b) 0 sin(b); 0 1 0; -sin(b) 0 cos(b)];

   % Find IA matrix, the IAC matrix, and its derivative
   IA = R2*IA0*R2';
   IAC = 2*IA + 2*mA*(d^2)*(eye(3) - bh2*bh2');
   IACd = 2*bd * (bh2x*IA - IA*bh2x);
```

```
    % Find the Euler sequence matrix and its derivative
    E = [sin(th(3))*sin(th(2)) cos(th(3)) 0;
         sin(th(2))*cos(th(3)) -sin(th(3)) 0;
         cos(th(2)) 0 1];
    Ed = [cos(th(3))*sin(th(2))*thd(3) + sin(th(3))*cos(th(2))*thd(2)
        -sin(th(3))*thd(3) 0;
         cos(th(2))*cos(th(3))*thd(2) - sin(th(2))*sin(th(3))*thd(3)
             -cos(th(3))*thd(3) 0;
         -sin(th(2))*thd(2) 0 0];

    % Find the partials of E with respect to theta
    dE1 = zeros(3);
    dE2 = [sin(th(3))*cos(th(2)) 0 0;
            cos(th(2))*cos(th(3)) 0 0;
            -sin(th(2)) 0 0];
    dE3 = [cos(th(3))*sin(th(2)) -sin(th(3)) 0;
            -sin(th(2))*sin(th(3)) -cos(th(3)) 0;
            0 0 0];

    % Assemble partials times thetadot column by column
    dEthd = [dE1*thd dE2*thd dE3*thd];

    % Compute IS and HS matricies
    IS = IB + IAC + Jw;
    HS = IB*E*thd + IAC*(E*thd + bd*bh2) + Jw*(E*thd + ad);

    % Compute the f vector
    f = [-E' * (IS*Ed*thd + IACd*(E*thd + bd*bh2) + IAC*bdd*bh2) +
        (dEthd - Ed)'*HS; u - Jw*Ed*thd];

    % Compute the M matrix
    M = [E'*IS*E E'*Jw; Jw*E Jw];

    % Find the angle accelerations and pass through the angular rates
    xd = [thd; ad; M\f];

end
```

Then the remaining code setup the simulation parameters, and ran the simulation using ode45:

Listing 2: Running the zero-torque simulation.

```
% Define the beta angle functions
bm   = deg2rad(60);
tm   = 60;
bt   = @(t)
    bm*(10*(t/tm).^3-15*(t/tm).^4+6*(t/tm).^5).*(t<=tm)+bm*(t>tm);
bdt  = @(t) (bm/tm)*(30*(t/tm).^2-60*(t/tm).^3 +
    30*(t/tm).^4).*(t<=tm);
bddt = @(t)
    (bm/tm^2)*(60*(t/tm)-180*(t/tm).^2+120*(t/tm).^3).*(t<=tm);

% Define the system constant parameters
IB  = diag([80 100  60]);
IA0 = diag([6  30   35]);
Jw  = diag([0.1 0.12 0.08]);
mA  = 24;
d   = 1.5;

% Define the initial condition of the simulation and a constant zero
    torque
x0 = [pi/3; pi/3; pi/3; zeros(6,1); 300; -200; 100];
u = zeros(3,1);

% Run the simulation
tspan = [0 100];
tol = 10^(-12);
opt = odeset('RelTol', tol, 'AbsTol', tol);
[simtime, xsim] = ode45(@(t,x) rwEoM(x, t, u, bt, bdt, bddt, IB, IA0,
    Jw, mA, d), tspan, x0, opt);
```

Then the following plots show the resulting time histories for $\boldsymbol{\theta}$, $\boldsymbol{\omega}$, and $\boldsymbol{\Omega}$. Note that to plot $\boldsymbol{\omega}$, the value $\dot{\boldsymbol{\theta}}$ had to be mapped to $\boldsymbol{\omega}$ by the relation:

$$\boldsymbol{\omega} = \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} \tag{80}$$

Which was given by the problem description.

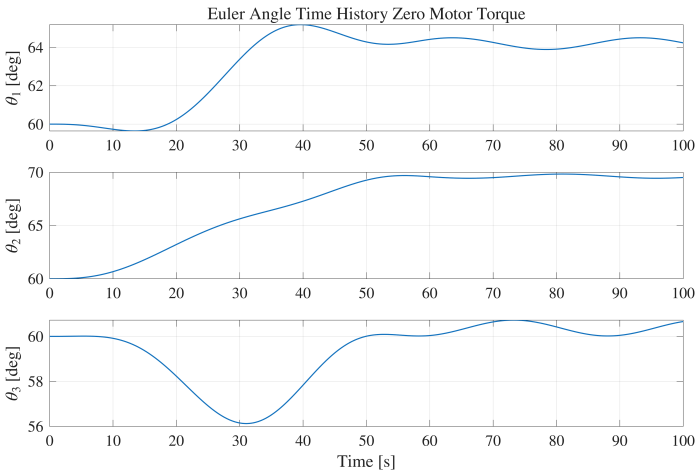



Figure 3: Time history of the Euler angles $\boldsymbol{\theta}$ with zero RW motor torque, $\boldsymbol{u} = \mathbf{0}$.

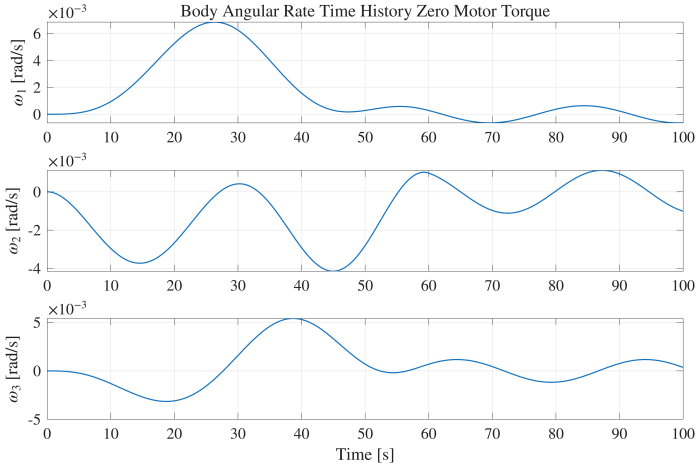



Figure 4: Time history of the body angular rate $\boldsymbol{\omega}$ with zero RW motor torque, $\boldsymbol{u} = \mathbf{0}$.

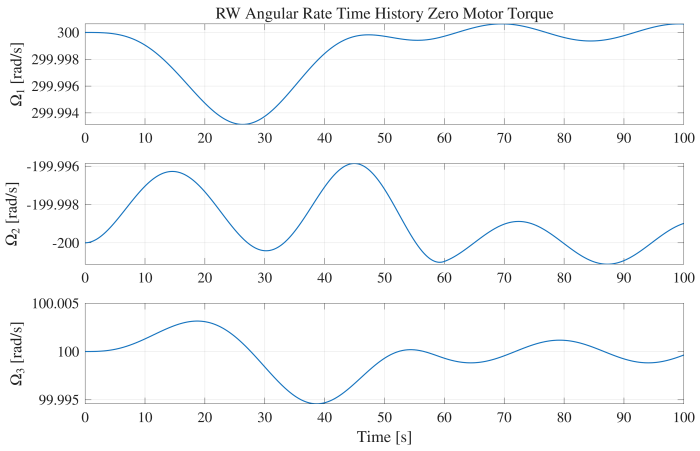



Figure 5: Time history of the RW spin rates relative to the bus $\boldsymbol{\Omega}$ with zero RW motor torque, $\boldsymbol{u} = \boldsymbol{0}$.

**(b.3): Find**

Plot the time history of $\boldsymbol{p_\alpha}$, and discuss its time variation and why the result makes sense. Also, discuss why $\boldsymbol{\Omega}(t)$ is generally not constant.
**Analysis/Conclusions:**

To plot the time history of $\boldsymbol{p_\alpha}$ we can use the analytical expression found for it in (a.5):

$$\boldsymbol{p}_\alpha = \boldsymbol{J}_w \left( \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\alpha}} \right) = \boldsymbol{J}_w \left( \boldsymbol{\omega} + \boldsymbol{\Omega} \right) \tag{81}$$

Then, implementing this expression into code and plotting its three vector values, we get the following result:

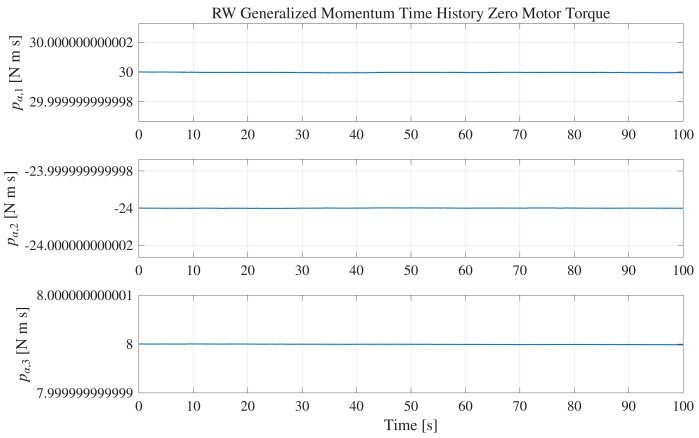



Figure 6: Time history of the generalized momentum $\boldsymbol{p_\alpha}$ with zero RW motor torque.

We see in the plot that $\boldsymbol{p_\alpha}$ **does not vary with time**.

This is the expected result because in (a.6) we found $\dot{\boldsymbol{p}}_{\boldsymbol{\alpha}} = \boldsymbol{u}$, and with $\boldsymbol{u} = \boldsymbol{0}$ there is no torque along the wheel axes to change it. The generalized momentum $\boldsymbol{p_\alpha} = \boldsymbol{J}_w (\boldsymbol{\omega} + \boldsymbol{\Omega})$ depends on the angular rate of each wheel with respect to the inertial frame, which combines both the body rotation and the wheel rate relative to the body. Likewise, even though $\boldsymbol{\Omega}$ changes over time, it only does so in response to the changes in $\boldsymbol{\omega}$ caused by the array maneuver, such that the sum $\boldsymbol{\omega}+\boldsymbol{\Omega}$, and therefore $\boldsymbol{p_\alpha}$, is maintained.

So, $\boldsymbol{\Omega}$ is generally not constant because any change in the angular rate of the body frame $\boldsymbol{\omega}$ must be offset by an equal and opposite change in the wheel rates, and in our scenario the arrays are usually causing some change in $\boldsymbol{\omega}$.

**(b.4): Find**

From the symbolic expression of $\boldsymbol{p_\alpha}$, derive the relation between $\boldsymbol{\Omega}(t)$ and $\boldsymbol{\omega}(t)$ under $\boldsymbol{u}(t) = 0$. Verify that this analytical relation between $\boldsymbol{\Omega}(t)$ and $\boldsymbol{\omega}(t)$ is consistent with the numerical result.

**Analysis/Conclusions:**

The symbolic expression for $\boldsymbol{p_\alpha}$ was derived in (a.5) as:

$$\boldsymbol{p}_\alpha = \boldsymbol{J}_w \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) \tag{82}$$

It was then found in (a.6) that the time derivative of $\boldsymbol{p_\alpha}$ is:

$$\dot{\boldsymbol{p}}_{\boldsymbol{\alpha}} = \boldsymbol{u} \tag{83}$$

when considering the fact that $\boldsymbol{\alpha}$ is cyclic. Thus this means that when $\boldsymbol{u} = 0$:

$$\dot{\boldsymbol{p}}_{\boldsymbol{\alpha}} = \boldsymbol{u} = 0 \tag{84}$$

so the value of $\boldsymbol{p_\alpha}$ must be constant.

So, for considering the relationship of $\boldsymbol{\omega}$ and $\boldsymbol{\Omega}$, we can move $\boldsymbol{J}_\omega$ to the other side, and consider $\boldsymbol{J}_w^{-1}\boldsymbol{p_\alpha}$ as a constant $C$:

$$\begin{aligned} \boldsymbol{p}_\alpha &= \boldsymbol{J}_w \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) \\ \boldsymbol{J}_w^{-1}\boldsymbol{p}_{\boldsymbol{\alpha}} &= \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) \\ C &= \left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) \\ \text{Where: } C &= \boldsymbol{J}_w^{-1}\boldsymbol{p}_{\boldsymbol{\alpha}} \end{aligned} \tag{85}$$

> So, since $C$ is always constant, the sum $\boldsymbol{\omega} + \boldsymbol{\Omega}$ must always be constant for all possible values of one or the other.

To find the value of $C$ one simply needs to add the initial conditions, which for the initial conditions in this part evaluates to:

$$C = \boldsymbol{\omega}(0) + \boldsymbol{\Omega}(0) = \boldsymbol{0} + [300, -200, 100]^\top = [300, -200, 100]^\top \tag{86}$$

Thus, to verify these values stay constant, we can simply plot the deviation of the sum $\boldsymbol{\omega} + \boldsymbol{\Omega}$ from $[300, -200, 100]^\top$ over the whole simulation. The resulting code plots this numerical verification:

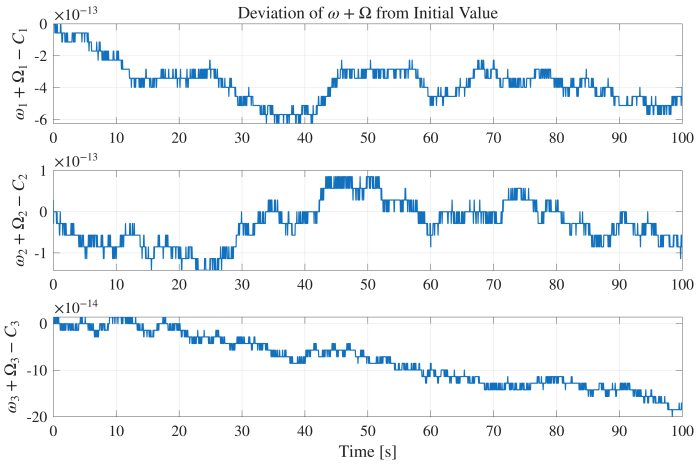



Figure 7: Deviation of $\boldsymbol{\omega} + \boldsymbol{\Omega}$ from $C = [300, -200, 100]^\top$ rad/s with zero RW motor torque.

We can see in the plots above that the deviation stays within the order of magnitude of about $10^{-13}$ which is well within the range of numerical error, **thus the analytical relation is consistent with the numerical result**.

## (c): Given

Let us then simulate the spacecraft attitude for an arbitrary RW torque of my choice.

### (c.1): Find

First, define and report the RW torque as a smooth function of time $\boldsymbol{u}(t)$, and plot its time history over $0 \leq t \leq 60$ sec.

**Analysis/Conclusions:**

It is my understanding that the stable axis of this spacecraft at any time should be the axis normal to the solar array. Likewise, I will define my torque profile $\boldsymbol{u}$ to have its vector direction always pointing in the direction of $\hat{\boldsymbol{n}}_A$ scaled by some magnitude $\|\boldsymbol{u}\|_2$ in an attempt to keep the rotation of the spacecraft stable as torque is applied.

Recall that the vector normal to the arrays is given by:

$$\hat{\boldsymbol{n}}_A = \boldsymbol{R}_2(\beta)\hat{\boldsymbol{b}}_3 = \begin{bmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \sin\beta \\ 0 \\ \cos\beta \end{bmatrix} \tag{87}$$

After some testing using the code from part (b) with different torque values, it was decided to have a $\|\boldsymbol{u}\|_2$ value of 1 N m. So, the torque profile is defined as:

$$\boldsymbol{u}(t) = \|\boldsymbol{u}\|_2 \, \hat{\boldsymbol{n}}_A(t) = \|\boldsymbol{u}\|_2 \begin{bmatrix} \sin\beta(t) \\ 0 \\ \cos\beta(t) \end{bmatrix}, \quad \text{Where: } \|\boldsymbol{u}\|_2 = 1 \text{ N m} \tag{88}$$

Then, after code implementation, the profile plots as follows:

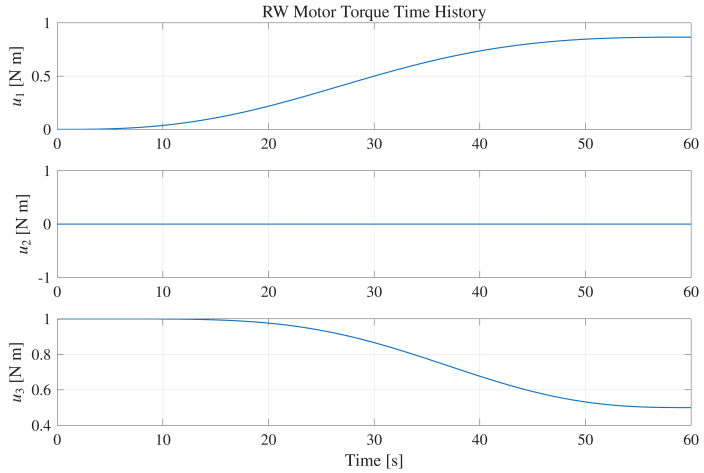



Figure 8: Time history of the RW motor torque $\boldsymbol{u}$ applied along the solar panel normal $\hat{\boldsymbol{n}}_A$.

**(c.2): Find**

Numerically propagate the system for the same time span using the EoM implemented above. Plot the time histories of $\boldsymbol{\theta}$, $\boldsymbol{\omega}$, $\boldsymbol{\Omega}$, and $\boldsymbol{p_\alpha}$. Discuss the result and why it makes sense.

**Analysis/Conclusions:**

Using the torque profile of $\boldsymbol{u}$ derived in (c.1), the same code for propagating the results in part (b) was ran with this control profile entered in as the argument for $\boldsymbol{u}$ over time. The resulting plots can be seen below:

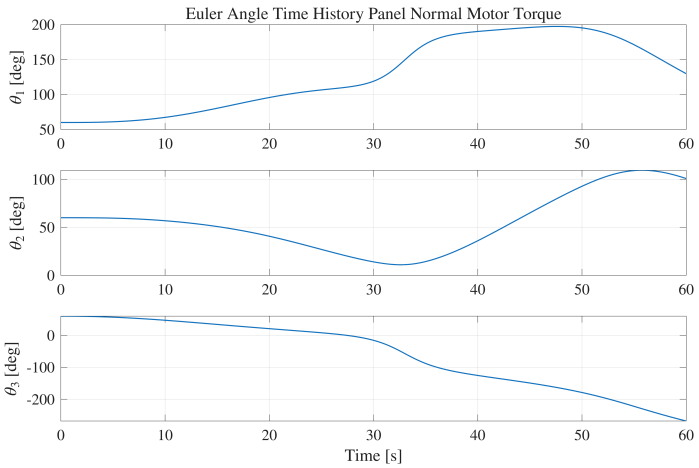



Figure 9: Time history of the Euler angles $\boldsymbol{\theta}$ with the RW motor torque applied along the solar panel normal.

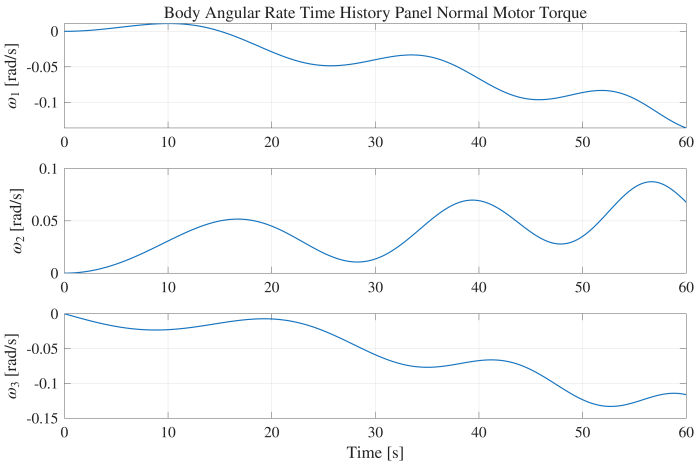



Figure 10: Time history of the body angular rate $\boldsymbol{\omega}$ with the RW motor torque applied along the solar panel normal.

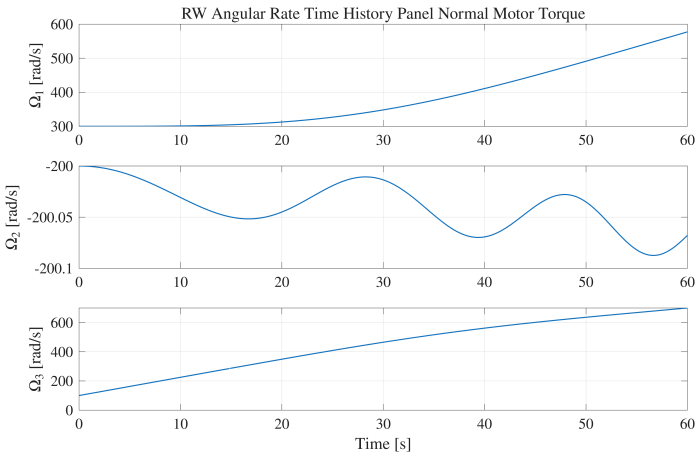



Figure 11: Time history of the RW spin rates relative to the bus $\boldsymbol{\Omega}$ with the RW motor torque applied along the solar panel normal.

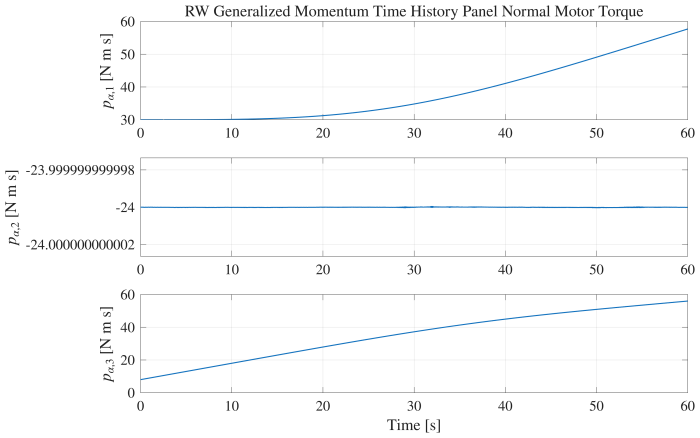



Figure 12: Time history of the generalized momentum $\boldsymbol{p}_\alpha$ with the RW motor torque applied along the solar panel normal.

The results of the $\boldsymbol{\Omega}$ and $\boldsymbol{p_\alpha}$ plots make sense because we see from the torque profile derivation:

$$\boldsymbol{u}(t) = \|\boldsymbol{u}\|_2 \, \hat{\boldsymbol{n}}_A(t) = \|\boldsymbol{u}\|_2 \begin{bmatrix} \sin\beta(t) \\ 0 \\ \cos\beta(t) \end{bmatrix}, \quad \text{Where: } \|\boldsymbol{u}\|_2 = 1 \text{ N m} \tag{89}$$

that torque is only applied to wheels 1 and 3. Likewise, we see in the $\boldsymbol{\Omega}$ and $\boldsymbol{p_\alpha}$ plots a continuous increase in rotor angular rate and generalized momentum on those axes respectively, where the slope of each $p_\alpha$ component matches its motor torque since $\dot{\boldsymbol{p}}_{\boldsymbol{\alpha}} = \boldsymbol{u}$. We see $p_{\alpha_1}$ steepen as $\sin\beta$ grows, while $p_{\alpha_3}$ flattens slightly as $\cos\beta$ decreases. Furthermore, we see the momentum on the second axis is constant, which is consistent with no motor torque applied on that axis since $\dot{p}_{\alpha_2} = u_2 = 0$. We see an oscillation in the $\Omega_2$ plot despite the constant momentum because the changing $\omega_2$ value forces an equal and opposite change in $\Omega_2$, since the sum $\boldsymbol{\omega} + \boldsymbol{\Omega}$ is maintained as discussed in (b.3).

Likewise, we see oscillations in $\omega_1$ and $\omega_3$, which would be expected since the spacecraft is not spinning in a symmetric manner, so its rotation couples between the axes. The Euler angles also change relatively slowly, which matches the magnitude of the angular rates in the $\boldsymbol{\omega}$ plots of about 0.1 rad/s.

## (d): Given/Find

Discuss a high-level sketch of the derivation procedure for the EoMs of this system using the Euler's rotational equations of motion, show the complete derivation, and verify its consistency with the EoM derived above.

**Analysis/Conclusions:**

It is known that Euler's rotational equations of motion can be written in tensor form as the following:

$$I_c\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I_c\boldsymbol{\omega}) = \boldsymbol{\tau}_c \tag{90}$$

Where:

- $I_c$ is the inertial tensor some rigid body.
- $\boldsymbol{\omega}$ is the angular rate of that rigid body with respect to the inertial frame.
- $\dot{\boldsymbol{\omega}}$ is the derivative of said angular rate.
- $\boldsymbol{\tau}_c$ is the external torques acting on the system.

It itself is derived from the following application of the transport theorem:

$$\dot{\boldsymbol{H}}_c = \frac{^{\mathcal{B}}d}{dt}\boldsymbol{H}_c + \boldsymbol{\omega} \times \boldsymbol{H}_c \tag{91}$$

Where:

- $\boldsymbol{H}_c$ is the total angular momentum of the rigid body.
- $\dot{\boldsymbol{H}}_c$ is the derivative of said angular momentum, which itself equals torque.
- $\frac{^{\mathcal{B}}d}{dt}\boldsymbol{H}_c$ is the body-frame relative derivative of $\boldsymbol{H}_c$.
- $\boldsymbol{\omega}$ once again is the angular rate of that rigid body with respect to the inertial frame.

From the derivations in part (a), it was already found that the total angular momentum of the spacecraft is:

$$\boldsymbol{H}_S = \boldsymbol{I}_B\boldsymbol{\omega} + \boldsymbol{I}_{A/C}\left(\boldsymbol{\omega} + \dot{\beta}\hat{\boldsymbol{b}}_2\right) + \boldsymbol{J}_w\left(\boldsymbol{\omega} + \boldsymbol{\Omega}\right) \tag{92}$$

Where:

- $\boldsymbol{\omega}$ represents the same quantity as in Euler's rotational equations.
- $\boldsymbol{\Omega}$ is the angular rate of the reaction wheel rotors.
- $\boldsymbol{I}_B$, $\boldsymbol{I}_{A/C}(t)$, and $\boldsymbol{J}_w$ are inertial tensor terms of different components.
- $\hat{\boldsymbol{b}}_2$ is the second body frame axis.

- $\dot{\beta}$ models the rate of change of the prescribed motion of the solar arrays.

So, we can leverage the relationships between the three above equations for the full derivation of the equations. Here are some key points:

Note all three equations use the same quantity $\boldsymbol{\omega}$ for the same value (angular rate of the body frame WRT inertial frame). This term can be used to introduce the Euler angle coordinates $\boldsymbol{\theta}$ which had their relationship to $\boldsymbol{\omega}$ defined in earlier parts as:

$$\boldsymbol{\omega} = \boldsymbol{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}, \qquad \boldsymbol{E}(\boldsymbol{\theta}) = \begin{bmatrix} \sin\theta_3 \sin\theta_2 & \cos\theta_3 & 0 \\ \sin\theta_2 \cos\theta_3 & -\sin\theta_3 & 0 \\ \cos\theta_2 & 0 & 1 \end{bmatrix} \tag{93}$$

Euler's rotational equations introduce an external torque term $\boldsymbol{\tau}_c$. In our model in earlier parts, the external torque came purely from the motor torque $\boldsymbol{u}$ which means we can substitute $\boldsymbol{\tau}_c$ for $\boldsymbol{u}$. As mentioned, the value of $\dot{\boldsymbol{H}}_S$ equals the torque $\boldsymbol{\tau}$ in Euler's equations, so we can say $\dot{\boldsymbol{H}}_S = \boldsymbol{u}$.

The usage of the transport theorem in Eq. (91), can be used to introduce $\boldsymbol{H}_S$ to the derivation. Likewise, the angular momentum term $\boldsymbol{H}_S$ introduces both the effect of the solar arrays and the reaction wheels and how they relate to Euler's rotational equations. We see that $\boldsymbol{\Omega} = \dot{\boldsymbol{\alpha}}$ is introduced here, and its derivative $\ddot{\boldsymbol{\alpha}}$ will itself will enter the derivation from derivatives of $\boldsymbol{H})_S$. Likewise, second derivative terms of $\boldsymbol{\theta}$ will also enter the equation here through its relationship with $\boldsymbol{\omega}$.

In the Lagrangian approach to this derivation, $\boldsymbol{\alpha}$ was never itself included due to it being cyclic, and we see the same thing will occur in this derivation due to $\boldsymbol{\alpha}$ only being introduced by $\boldsymbol{H}_S$ as its higher order derivative $\boldsymbol{\Omega}$.

These key points show that this derivation covers the bases and gives a high overview on how the essential terms for the EoM enter the derivation.

Likewise, the high level sketch of the derivation would be as follows:

- Use the transport theorem to write Euler's rotational equations in terms of the total spacecraft angular momentum $\boldsymbol{H}_S$, and introduce motor torque $\boldsymbol{u}$ via $\dot{\boldsymbol{H}}_S$.
- Take the body-frame derivative of $\boldsymbol{H}_S$ which importantly brings in $\dot{\boldsymbol{\alpha}}$ and $\ddot{\boldsymbol{\alpha}}$.
- Substitute $\boldsymbol{\omega} = \boldsymbol{E}\dot{\boldsymbol{\theta}}$ and $\dot{\boldsymbol{\omega}} = \dot{\boldsymbol{E}}\dot{\boldsymbol{\theta}} + \boldsymbol{E}\ddot{\boldsymbol{\theta}}$ to also introduce $\boldsymbol{\theta}$, $\dot{\boldsymbol{\theta}}$, and $\ddot{\boldsymbol{\theta}}$.
- Evaluate all remaining terms like the cross product and algebraically separate $\boldsymbol{q}$, $\dot{\boldsymbol{q}}$, and $\ddot{\boldsymbol{q}}$ terms.

- Collect the $\ddot{\boldsymbol{q}}$ terms into matrix form $\boldsymbol{M}$ and the $\dot{\boldsymbol{q}}$ and $\boldsymbol{q}$ terms into the vector function $\boldsymbol{f}$ to finish the derivation.